

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System for the Municipality of Oton

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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System Development Track

Second Semester, Academic Year 2025-2026

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The agricultural sector remains a cornerstone of the Philippine economy, serving as a primary engine for rural livelihood and economic sustenance across the archipelago. Within this landscape, livestock farming emerges as a critical pillar for ensuring national food security, augmenting household incomes, and catalyzing localized development. In the municipality of Oton, Iloilo, cattle raising is established as a vital agricultural tradition, contributing significantly to the provincial meat supply and livestock production chain. For the local farming community, cattle assets represent more than just a consistent revenue stream; they function as high-value long-term investments that provide essential financial resilience. The escalating market demand for animal products underscores the urgent necessity for optimized management frameworks to ensure the sustainability and profitability of these agricultural operations.

Data from the Philippine Statistics Authority (PSA, 2023) indicates that the livestock sub-sector is a dominant force in the nation's agricultural output. Specialized breeds, including Brahman, Holstein, and indigenous Bisaya crossbreeds, are predominantly managed due to their biological adaptability and superior economic performance. These assets are strategically utilized for genetic breeding, meat processing, and dairy production, offering raisers a viable pathway toward improved economic status. However, maximizing these biological yields

necessitates rigorous surveillance of reproductive cycles, pathological conditions, immunization timelines, and comprehensive herd synchronization.

Despite the immense economic value generated by the cattle industry in Oton, Iloilo, current operational paradigms continue to rely on antiquated manual recording and monitoring methodologies. The existing livestock insemination and breeding workflows are largely facilitated by field technicians under the Municipal Agriculture Office. These personnel manage a complex array of responsibilities vital to herd productivity, including the surveillance of artificial insemination windows, estrus cycle tracking, rebreeding coordination for failed cycles, pregnancy monitoring, vaccination logistics, and the documentation of mortality and health metrics.

Currently, these critical datasets are manually encoded into paper-based logbooks and physical ledgers. This traditional approach, while historically persistent, introduces severe operational bottlenecks that compromise the integrity and efficiency of municipal livestock management. Physical records are highly susceptible to environmental degradation from moisture, thermal exposure, fire, and biological pests. Furthermore, manual systems are prone to misplacement and physical loss, while the retrieval and updating of records are characterized by significant labor inefficiencies. As livestock inventories expand, the cumulative volume of paper records makes the retrieval of historical breeding patterns and the generation of administrative reports an increasingly arduous and error-prone endeavor.

The Food and Agriculture Organization (FAO, 2021) asserts that digital information management systems offer a superior and more resilient framework for agricultural data, enhancing accessibility and long-term preservation. In contrast to manual methods, digital infrastructures allow for the secure storage of high-density datasets while enabling real-time data retrieval and updates. Implementing such technological capabilities is essential for enhancing the operational throughput of local agricultural agencies and specialized livestock programs.

A secondary challenge involves the persistent communication gap and lack of synchronized information transit between producers and agricultural personnel. A significant portion of the livestock raisers in rural sectors belong to older demographics with limited digital literacy, relying on legacy communication channels such as personal visits, phone calls, or SMS. This decentralization often results in the delayed reporting of critical events, such as parturition, disease outbreaks, animal mortality, and vaccination needs. Consequently, livestock technicians may only receive field data days or weeks after an occurrence, hindering the synchronization of municipal databases.

These communication barriers introduce profound inaccuracies into the livestock management lifecycle. Technicians are frequently forced to utilize approximations for insemination dates, pregnancy durations, and estrus cycles due to fragmented field reporting. These estimations degrade the quality of official records, potentially reducing reproductive success rates and skewing municipal statistics. Moreover, delayed reporting impedes the deployment of

time-sensitive veterinary interventions and epidemiological surveillance, heightening the risk of contagion and financial loss within farming clusters.

The persistent reliance on manual documentation and fragmented communication ultimately degrades the analytical quality of reports submitted by the Municipal Agriculture Office to provincial oversight bodies. High-fidelity livestock data is paramount for strategic planning, resource distribution, disease monitoring, and evidence-based policy formulation. As emphasized by the World Bank (2019), the integration of digital agricultural technologies is essential for improving sector productivity through enhanced data accuracy and streamlined stakeholder coordination.

The rapid evolution of information and communication technology provides a transformative opportunity to address these systemic vulnerabilities. Web and mobile-based infrastructures offer innovative frameworks for managing agricultural data and facilitating real-time stakeholder interaction. These platforms enable automated notifications, centralized data repositories, and precision monitoring of field activities. By automating these workflows, digital systems reduce administrative burdens, minimize human error, and ensure the consistent availability of actionable information for municipal livestock management.

In response to these challenges, this study proposes the development of **BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System**. This platform

is engineered to modernize existing manual workflows by providing a centralized, user-friendly interface for technicians, raisers, and administrators. By leveraging distributed technologies, BreedSmart seeks to automate breeding cycle computations using breed-specific reproductive logic, facilitating precise surveillance of insemination schedules, expected delivery dates, and overall reproductive efficiency.

Beyond reproductive monitoring, the system implements a robust digital framework for comprehensive health surveillance. Features including vaccination tracking, disease reporting, clinical assessments, and mortality logging enable the maintenance of an organized digital registry. The cloud-synchronized database ensures that these records remain secure, auditable, and readily accessible for administrative reporting and data-driven decision-making.

The system further aims to bridge the communication gap between producers and the Municipal Agriculture Office. Through the mobile client, farmers can electronically submit service requests, report calving events, and notify technicians of health concerns while receiving automated alerts regarding breeding milestones and immunization windows. These features are designed to eliminate communication latency and improve field data fidelity. Simultaneously, technicians and office administrators gain real-time access to synchronized records, facilitating more effective service delivery and operational oversight.

Additionally, BreedSmart is anticipated to optimize reporting efficiency within the municipal agricultural sector. By generating standardized digital datasets, the platform streamlines the

preparation of analytical reports for provincial and national agencies. This capability enhances the reliability of regional livestock statistics, supports agricultural policy development, and ensures the efficient implementation of government programs targeted at local farming communities.

Ultimately, the realization of **BreedSmart** marks a pivotal transition toward the digitalization of livestock management in Oton, Iloilo. By replacing vulnerable manual processes with a synchronized digital ecosystem, the study aims to strengthen record integrity, enhance stakeholder coordination, and promote evidence-based agricultural management. These systemic improvements are intended to support the long-term competitiveness and sustainability of the local cattle industry, aligning with broader objectives of regional food security and agricultural modernization.

1.2 Statement of the Problem

The central difficulty addressed by this study lies in the persistent inefficiencies, data inconsistencies, and communication barriers experienced in livestock breeding and health management within the municipality of Oton, Iloilo. Although cattle farming remains an essential component of the local agricultural economy, the operational systems supporting livestock monitoring continue to rely heavily on manual and fragmented processes. These traditional methods include paper-based recording, phone call coordination, handwritten scheduling, and delayed reporting mechanisms, which, while familiar to farmers and livestock personnel, are inefficient, error-prone, and difficult to sustain in modern agricultural operations. The following subsections elaborate on the specific problems, their underlying causes, and their impacts on farmers, livestock technicians, administrators, and the broader agricultural modernization agenda.

Data Vulnerability

Data vulnerability manifests in the continued reliance on paper-based logs and handwritten livestock records. In Oton, livestock technicians manually document breeding schedules, insemination records, vaccination histories, mortality reports, and pregnancy monitoring information using notebooks and physical ledgers. Although these methods have long been practiced within local agricultural offices, they are highly susceptible to physical deterioration, accidental loss, and data misplacement. According to the Food and Agriculture Organization (FAO, 2021), manual agricultural documentation systems reduce operational efficiency and increase the risk of inaccurate or incomplete records.

The root cause of this vulnerability lies in the absence of a centralized digital storage system capable of preserving and organizing livestock information in real-time. Paper-based records are difficult to retrieve and analyze, especially when technicians need historical breeding data for reporting and monitoring purposes. In many cases, environmental conditions such as moisture, heat exposure, and mishandling contribute to the degradation of physical documents. For livestock technicians managing multiple farmers across rural barangays, locating and verifying records becomes increasingly time-consuming and inefficient.

The implications of data vulnerability extend beyond administrative inconvenience. Missing or damaged records compromise the accuracy of livestock monitoring programs, vaccination scheduling, and breeding management. For example, a technician may fail to verify a cow's previous insemination history due to incomplete documentation, leading to delays in reproductive monitoring and decreased breeding efficiency. Such inefficiencies reduce the reliability of livestock data submitted to municipal and provincial agricultural offices, ultimately affecting agricultural planning and resource allocation.

Inaccurate Livestock Reporting

Inaccurate livestock reporting further compounds the challenges experienced within the municipality's livestock sector. Many cattle farmers in Oton rely primarily on phone calls or text messages to report important livestock events such as calf births, failed inseminations, animal illnesses, or deaths. Due to delayed communication and the absence of real-time reporting systems, livestock technicians are frequently forced to estimate breeding schedules, calf drop dates, and cattle population figures.

This issue aligns with findings from the World Bank (2019), which emphasized that unreliable agricultural data weakens decision-making processes and reduces the effectiveness of agricultural programs. The lack of synchronized digital reporting systems creates discrepancies between field records and official government submissions. In situations where farmers fail to immediately report successful births or animal health conditions, technicians often rely on assumptions and approximations when preparing reports for the Provincial Agriculture Office.

The consequences of inaccurate reporting affect multiple stakeholders. For farmers, delayed monitoring may reduce breeding productivity and increase the likelihood of reproductive complications. For technicians, inaccurate reports create additional workloads and reduce confidence in livestock statistics. At the administrative level, unreliable data compromises agricultural policy planning, disease monitoring programs, and livestock population assessments. In the long term, the absence of accurate reporting mechanisms weakens the municipality's capacity to implement evidence-based agricultural development initiatives.

Communication Gap

The communication gap between farmers and livestock technicians represents another significant problem within the current livestock management process. At present, there is no centralized platform where farmers can directly request insemination services, report animal diseases, or receive breeding reminders in real-time. Communication largely depends on informal channels such as phone calls, text messaging, and personal visits to the municipal agriculture office.

According to the International Livestock Research Institute (ILRI, 2020), effective digital communication systems are essential in improving livestock productivity and ensuring timely agricultural service delivery. However, in Oton, the absence of a unified communication platform limits coordination between farmers and agricultural personnel. Delays in communication often result in missed insemination schedules, late disease reporting, and inefficient service response times.

For example, a farmer from a remote barangay may experience delays in reporting a cow's reheat cycle due to the inability to immediately contact the livestock technician. Since cattle breeding schedules are highly time-sensitive, such delays may result in missed insemination opportunities and reduced reproductive success rates. Similarly, disease outbreaks may spread more rapidly when animal health concerns are not reported promptly to agricultural authorities.

The communication gap also contributes to the exclusion of critical livestock information from official records. Because communication is decentralized and inconsistent, many livestock events remain undocumented or are reported too late to be accurately incorporated into municipal databases. This lack of coordination weakens the efficiency of livestock monitoring operations and creates barriers to effective agricultural management.

Management Complexity

Management complexity represents another operational challenge in the municipality's livestock sector. Different cattle breeds such as Brahman, Holstein, and local Bisaya cattle possess varying reproductive cycles, maturity timelines, and breeding requirements. Despite these biological differences, the current system relies entirely on manual calculations and technician experience to determine breeding schedules and monitor reproductive progress.

The World Organisation for Animal Health (WOAH, 2021) emphasized that digital livestock monitoring systems improve reproductive management and reduce human error in animal health tracking. However, without automated breed-specific scheduling features, livestock technicians in Oton must manually estimate insemination intervals, pregnancy timelines, and expected delivery dates for each animal.

This management process becomes increasingly difficult when a single technician handles numerous farmers and livestock records simultaneously. Manual tracking increases the possibility of scheduling errors, overlapping appointments, and missed breeding opportunities. For example, a technician managing multiple cattle breeds may incorrectly calculate the insemination schedule of a Holstein cow using the reproductive timeline of a Brahman breed, leading to reduced breeding efficiency and financial losses for the farmer.

The complexity of livestock management not only affects reproductive monitoring but also limits the technician's ability to provide timely and accurate agricultural services. As livestock populations continue to grow, the limitations of manual monitoring systems become more evident, highlighting the need for automated and technology-driven solutions.

Data Inconsistency and Lack of Synchronization

Data inconsistency remains a persistent issue due to the absence of a synchronized livestock management system connecting field operations and administrative offices. Livestock records collected from farmers are often incomplete, duplicated, or delayed before reaching the municipal database. This fragmented process creates inconsistencies between actual field conditions and official agricultural records.

According to the Department of Agriculture (2022), modern agricultural systems require integrated digital infrastructures to improve data reliability, transparency, and operational efficiency. However, the current livestock management process in Oton lacks centralized synchronization capabilities, making it difficult for administrators to generate accurate livestock statistics and monitor long-term agricultural trends.

The absence of synchronized systems affects disease surveillance, breeding evaluations, vaccination programs, and livestock inventory management. In many cases, livestock information gathered during field visits remains isolated within physical logbooks and is not immediately reflected in administrative records. This disconnect limits the municipality's ability to produce timely reports and implement data-driven agricultural interventions.

Furthermore, inconsistent livestock data weakens the municipality's efforts toward agricultural modernization and digital transformation. Without reliable digital infrastructures, Oton risks falling behind in adopting smart agricultural technologies that could improve productivity, sustainability, and livestock management efficiency.

Implications

The implications of these problems are multifaceted. For farmers, inefficient livestock monitoring systems lead to delayed services, reduced breeding success, financial losses, and limited access to timely agricultural support. For livestock technicians, manual processes increase

workloads, reduce operational efficiency, and heighten the risk of human error. For administrators, inconsistent and unreliable data weakens agricultural planning, reporting accuracy, and policy implementation.

At the municipal level, the absence of a centralized digital livestock management platform undermines efforts toward agricultural modernization and sustainable rural development. Addressing these problems is therefore not only a technological necessity but also an economic and administrative imperative for the municipality of Oton, Iloilo.

In response to these challenges, this study proposes BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System. The proposed system seeks to provide centralized livestock monitoring, automated breeding schedule calculations, real-time reporting capabilities, and synchronized digital records for farmers, technicians, and administrators. Through this system, the study aims to improve operational efficiency, strengthen communication, enhance reporting accuracy, and support the modernization of livestock management practices within the municipality.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System for the Municipality of Oton, Iloilo. Specifically, the study aims to improve the efficiency and accuracy of livestock record management, automate cattle breeding cycle monitoring, and strengthen communication between farmers, livestock technicians, and the Municipal Agriculture Office through the implementation of a centralized digital platform.

1.3.2 Specific Objectives

To achieve this general objective, the study pursues several specific objectives, each of which contributes to the overall efficiency, usability, and reliability of the proposed livestock management system.

- 1. Analyze the current challenges faced by livestock technicians and cattle farmers regarding manual livestock monitoring, breeding management, and record-keeping processes.**

This objective ensures that the proposed system is grounded in the actual operational realities experienced within the Municipality of Oton. By examining the existing manual practices used by farmers and livestock technicians, the study identifies key problems such as data vulnerability, delayed reporting, communication gaps, and inaccurate livestock records. Understanding these challenges provides the foundation for designing a system that addresses local agricultural needs rather than implementing generalized technological solutions that may not fit the municipality's operational environment.

- 2. Design a user-friendly and responsive web and mobile interface that allows farmers to request insemination services, report animal diseases, and monitor livestock activities efficiently.**

User interface (UI) and user experience (UX) design are essential factors influencing technology adoption, especially among rural users with varying levels of digital literacy. A responsive and accessible interface ensures that both farmers and agricultural personnel can effectively use the system across different devices, particularly smartphones, which are the most accessible digital tools in rural communities. By providing simplified navigation and organized features, the

system enables users to efficiently manage livestock-related activities without requiring advanced technical knowledge.

- 3. Develop an automated breeding cycle tracking module capable of calculating reheating schedules, insemination intervals, pregnancy timelines, and expected calf drop dates based on breed-specific reproductive logic.**

Livestock breeding management involves complex reproductive timelines that vary according to cattle breed characteristics. This objective addresses the limitations of manual calculations currently performed by livestock technicians. By automating breeding cycle tracking, the system minimizes human error and improves monitoring accuracy for breeds such as Brahman, Holstein, and local Bisaya cattle. Automated scheduling also enhances breeding efficiency by ensuring timely insemination procedures and accurate reproductive monitoring.

- 4. Implement a real-time notification and reminder system that alerts farmers and livestock technicians regarding veterinary schedules, vaccination activities, breeding milestones, and animal health concerns.**

Timely communication is essential in livestock management because breeding schedules, disease reporting, and vaccination programs are highly time-sensitive. This objective introduces automation features that provide instant notifications and reminders to users regarding important livestock events. Through automated alerts, the system reduces missed schedules, improves response times, and strengthens coordination between farmers and agricultural personnel. The notification system also minimizes the risk of delayed reporting and enhances overall livestock monitoring efficiency.

- 5. Create a centralized administrative dashboard for the Municipal Agriculture Office to monitor livestock records, manage farmer requests, and generate accurate reports for submission to the Provincial Agriculture Office.**

Administrative dashboards are essential tools for efficient agricultural management and decision-making. This objective focuses on providing the Municipal Agriculture Office with a centralized platform capable of organizing livestock data, monitoring field activities, and generating official reports. Through digital dashboards, administrators can easily access livestock statistics, breeding records, disease reports, and vaccination histories. The dashboard also improves data synchronization between field operations and administrative offices, reducing inconsistencies and improving report reliability.

- 6. Store and manage comprehensive livestock records, including animal profiles, breeding histories, vaccination records, disease monitoring data, and mortality reports using a secure database management system.**

Effective data management is critical for ensuring the reliability, accessibility, and security of livestock information. This objective emphasizes the development of a centralized database capable of securely storing and organizing large volumes of agricultural data. By maintaining digital livestock records, the system minimizes the risk of data loss associated with paper-based documentation. Secure database practices also protect sensitive agricultural information while improving long-term record accessibility for technicians and administrators.

- 7. Improve communication and coordination between farmers, livestock technicians, and agricultural administrators through a centralized digital platform.**

Communication gaps remain one of the major challenges in the current livestock management process within the municipality. This objective seeks to establish a unified communication

system where farmers can directly submit requests, report animal conditions, and receive updates from agricultural personnel in real-time. Improved coordination reduces delays in service delivery, enhances livestock monitoring, and strengthens the relationship between farmers and the Municipal Agriculture Office.

8. Test and evaluate the system's functionality, usability, reliability, responsiveness, and data accuracy within an actual agricultural management environment.

System evaluation is essential to determine whether the proposed platform effectively addresses the identified problems within the livestock sector. This objective focuses on assessing the system's performance using software quality standards and usability testing methods. Through evaluations involving livestock technicians and farmers, the study determines whether BreedSmart improves operational efficiency, reduces manual recording errors, and enhances livestock management processes within the Municipality of Oton, Iloilo.

1.4 Significance of the Study

The proposed system is expected to provide significant benefits to various stakeholders involved in livestock management, agricultural operations, academic development, and future technological research. By introducing a centralized web and mobile-based livestock insemination and health management system, the study aims to improve operational efficiency, communication, and data reliability within the Municipality of Oton, Iloilo. The following subsections discuss the significance of the study to specific groups and sectors.

Farmers

The proposed BreedSmart system will greatly benefit cattle farmers by providing a direct, accessible platform for requesting livestock-related services and monitoring animal health activities. Through the mobile application, farmers can submit insemination requests, report

animal diseases, monitor breeding schedules, and receive automated reminders regarding vaccination schedules, reheat cycles, and expected calf drop dates.

This functionality reduces the dependence on traditional communication methods such as phone calls, handwritten records, and physical visits to the Municipal Agriculture Office. By improving access to livestock information and agricultural services, the system enables farmers to make more informed decisions regarding breeding management and animal health monitoring. Furthermore, automated reminders help farmers avoid missed schedules, thereby improving breeding efficiency, reducing livestock health risks, and increasing overall productivity within their farming operations.

Livestock Technicians

The study will also provide substantial benefits to livestock technicians responsible for monitoring breeding activities and maintaining livestock records within the municipality. Currently, technicians rely heavily on manual calculations, handwritten logbooks, and fragmented communication methods, which consume considerable time and increase the risk of human error.

Through BreedSmart, livestock technicians will gain access to automated breeding cycle calculations, centralized livestock records, and real-time communication tools. The system minimizes the need for manual estimation of reheat cycles, insemination schedules, and calf drop dates by automatically generating breeding timelines based on breed-specific reproductive logic. This automation reduces administrative burdens, improves monitoring accuracy, and allows technicians to focus more on delivering agricultural services rather than managing paperwork.

Additionally, centralized digital records improve data accessibility and retrieval, enabling technicians to efficiently track livestock histories, vaccination records, mortality reports, and

disease monitoring activities. The system also enhances coordination between technicians and farmers by providing faster and more organized communication channels.

Oton Agriculture Office

The Municipal Agriculture Office of Oton will significantly benefit from implementing a centralized digital livestock management system. One of the primary challenges currently faced by the office involves inconsistent livestock records, delayed reporting, and fragmented field data. BreedSmart addresses these issues by providing synchronized and organized digital records accessible to authorized agricultural personnel.

The administrative dashboard enables the Agriculture Office to monitor livestock activities, manage farmer requests, and generate accurate reports for submission to the Provincial Agriculture Office. Through centralized data management, the office can improve the reliability, transparency, and accuracy of agricultural records. This capability strengthens decision-making processes, enhances livestock monitoring programs, and supports data-driven agricultural planning within the municipality.

Furthermore, the digitalization of livestock management operations contributes to the modernization of local agricultural services and aligns with broader government initiatives promoting digital transformation within rural communities.

BSIT Students

The study is also significant to Bachelor of Science in Information Technology (BSIT) students, particularly those interested in software development, database management, mobile applications, and agricultural technologies. BreedSmart serves as a practical application of information technology concepts within a real-world agricultural setting.

The development of the system demonstrates the integration of web development, mobile programming, database management systems, automation logic, and user interface design in

solving actual community problems. Through this study, BSIT students may gain insights into the role of technology in agricultural modernization and digital transformation. The project also serves as a reference for future capstone projects involving smart systems, automation, and government service applications.

Additionally, the study highlights the importance of developing socially relevant technologies that address operational challenges within local communities, encouraging students to create innovative solutions that contribute to public service improvement and rural development.

Future Researchers

Future researchers may benefit from this study as a reference for further investigations into digital agriculture, livestock management systems, and agricultural technological innovations. The findings, methodologies, system architecture, and implementation strategies presented in this research may serve as valuable resources for scholars conducting studies involving agricultural automation, mobile information systems, database integration, and smart farming technologies.

The study may also provide a foundation for future enhancements such as integrating Internet of Things (IoT) devices, artificial intelligence-based livestock monitoring, predictive analytics, geographic information systems (GIS), and cloud-based agricultural management platforms. Researchers may expand upon the proposed system to develop more advanced technological solutions that further improve livestock productivity, disease monitoring, and agricultural sustainability.

Ultimately, this study contributes to the growing body of knowledge regarding digital transformation within local government agricultural units and demonstrates the potential of information technology in improving agricultural operations, livestock management efficiency, and rural community development.

1.5 Theoretical Framework

System Development Life Cycle (SDLC)

The System Development Life Cycle (SDLC) is a structured process used in the development of information systems, which involves stages such as planning, analysis, design, development, testing, implementation, and maintenance. This model ensures that systems are developed in an organized and systematic way, reducing errors and improving overall system quality.

In the context of BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System, SDLC serves as the main development framework. The study follows each phase to ensure that the system is properly designed and aligned with the actual needs of livestock technicians and cattle farmers in the Municipality of Oton, Iloilo. During the planning and analysis phase, the study identifies problems such as manual record-keeping, inaccurate reporting, and communication gaps. The design and development phases focus on building the system features such as livestock tracking, breed-specific calculations, notification systems, and centralized record management.

Through SDLC, the system is developed in a structured and reliable manner, ensuring that the final output is functional, efficient, and suitable for real-world agricultural use.

Agile Methodology

Agile Methodology is a software development approach that focuses on iterative development, continuous feedback, and flexibility throughout the development process. It allows developers to improve and adjust the system based on user needs and testing results.

In this study, Agile is applied to ensure that BreedSmart is continuously improved during development. The system is built in small increments, allowing features such as insemination tracking, reheat cycle computation, and livestock record management to be tested and refined step by step. Feedback from potential users such as livestock technicians and farmers is considered during each development cycle to ensure that the system meets their actual needs.

Agile is important in this study because livestock management requirements may change depending on field conditions, making flexibility and adaptability essential in system development.

Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM), developed by Davis (1989), explains how users decide to accept and use a new technology based on perceived usefulness and perceived ease of use. Perceived usefulness refers to how much a user believes that the system improves their work, while perceived ease of use refers to how easy the system is to learn and operate.

In this study, TAM is relevant because the success of BreedSmart depends on whether farmers, livestock technicians, and agricultural staff are willing to use the system. If users find that the system helps them manage livestock records faster, reduces manual errors, and improves communication, they are more likely to adopt it. Likewise, if the system is simple, easy to navigate, and accessible through mobile devices, users will find it easier to use in their daily livestock management activities.

To support TAM, the system is designed with a simple interface, clear navigation, and mobile accessibility to ensure ease of use and practical usefulness in real farming environments.

Information Systems Success Model

The Information Systems Success Model by DeLone and McLean evaluates system success based on system quality, information quality, service quality, use, user satisfaction, and net

benefits. This model is widely used in assessing the effectiveness of information systems in real-world applications.

In this study, the model is used to evaluate the effectiveness of BreedSmart after development. System quality is measured through performance, usability, and reliability of the platform. Information quality is assessed through the accuracy of livestock records, breeding calculations, and generated reports. User satisfaction is evaluated based on feedback from farmers and livestock technicians after system use.

This framework ensures that the developed system is not only functional but also effective in improving livestock management processes in the Municipality of Oton, Iloilo.

1.6 Conceptual Framework

The conceptual framework of this study illustrates the overall structure of how the proposed system, **BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System**, is developed and how it transforms input data into a functional output. The study follows the Input–Process–Output (IPO) model, which is commonly used in system development research to clearly present the flow of information and system operation.

Input

The input phase includes all data and information gathered for the development of the system. This consists of the current manual livestock management practices in the Municipality of Oton, Iloilo, including paper-based records, breeding logs, vaccination records, and livestock monitoring reports. It also includes user requirements collected from livestock technicians and cattle farmers through interviews, observations, and surveys. These inputs help identify

common problems such as inaccurate records, delayed communication, and difficulties in tracking breed-specific breeding cycles.

Process

The process phase involves the actual development of the system using the System Development Life Cycle (SDLC) and Agile methodology. During this phase, the system is designed, developed, tested, and refined based on user requirements. Key processes include designing the system interface, developing the database, implementing breed-specific algorithms, and integrating notification and communication features.

The system also processes livestock data such as insemination records, reheat cycles, pregnancy timelines, and calf drop schedules. Through automated computation and real-time data processing, the system improves the accuracy and efficiency of livestock monitoring and management.

Output

The output of the study is a fully functional **Web and Mobile-Based Livestock Insemination and Health Management System (BreedSmart)**. The system provides centralized livestock record management, automated breeding cycle tracking, real-time communication between farmers and livestock technicians, and accurate reporting for the Municipal Agriculture Office.

The final output aims to reduce manual errors, improve data accuracy, enhance communication, and modernize livestock management practices within the Municipality of Oton, Iloilo.

1.7 Scope and Delimitation

1.7.1 Detailed Functional Capabilities and Operational Boundaries of System Modules (The Scope)

The primary focus of this study is the comprehensive design, architectural development, field implementation, and empirical software quality evaluation of **BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System**. The operational boundaries, processing modules, and data transit vectors of the software application are systematically organized into distinct functional layers explicitly assigned to three authorized user roles: Farmers (Clients), Livestock Technicians, and Administrators (Municipal Agriculture Office Personnel). Each subsystem is custom-tailored to address specific manual bottlenecks identified within the local government unit's current agricultural workflow.

The Farmer/Client Module is an interface engineered specifically to provide local livestock raisers with a highly responsive, user-friendly mobile application client designed to operate efficiently under volatile cellular network conditions in rural agricultural areas. Within this client module, the processing capabilities, user forms, and data transmission parameters are tightly limited to the following interactions:

- **User Profile and Account Management Subsystem:** Self-registration pipelines, security credential coordination, and digital profile updates, allowing farmers to safely manage their personal contact data, coordinate secure password updates, and pin their physical farm or barangay locations within designated municipal borders.
- **Animal Asset Digital Registry and Identity Module:** Interactive profile creation and inventory management for individual livestock assets, enabling users to digitally log

unique physical ear-tag values, specific breed profiles, biological gender, approximate birthdates, body condition scores, current status labels, and maternal lineage associations via parent record mapping (motherId) to preserve genetic tracking.

- **Veterinary Service Request Portal and Ticketing Interface:** Electronic submission of real-time field service requests for artificial insemination or urgent health checks, permitting farmers to describe localized symptoms using text strings, specify preferred calendar windows for field visits, define problem urgency levels, and attach supporting image uploads for remote appraisal by agricultural technicians prior to deployment.
- **Read-Only Information Retrieval Dashboard:** Direct, read-only dashboard access to historical individual health treatment ledgers, vaccination logs, deworming history, medical notes, automated breeding cycle notifications, upcoming appointment alerts, and scheduled veterinary reminders.

The Livestock Technician Module acts as a cross-platform mobile and web client designed for field personnel that prioritizes streamlined, rapid data entry to completely eliminate the vulnerabilities, physical degradation, and inaccuracies of manual, paper-based logbooks. The technical capabilities, input windows, and processing limits are scoped to the following actions:

- **Field Request Management and Dispatch Submodule:** Real-time mobile access, database-driven validation, status modification, and calendar scheduling approval of pending farmer insemination or veterinary service requests submitted from remote barangays.
- **Field Operations Data Logging and Treatment Engine:** Direct digital entry engines for recording artificial insemination procedures, outcome metrics like successful or failed fertilization attempts, precise sire breeds used, sire code tracking, specific vaccination

dosages, deworming dates, diagnoses, treatment notes, and field notes containing localized geospatial data and images captured during the visit.

- **Reproductive Surveillance and Lifecycle Monitoring Tool:** Updating active pregnancy diagnostic statuses following field check windows, maintaining reproductive logs, and recording physical calving events, including exact birth dates, the number of offspring delivered, sire details, and historical parity tracking for individual dams.

The Administrative Dashboard Module provides a web-based software platform designed for the central staff at the Municipal Agriculture Office that concentrates on large-scale data aggregation, multi-tier synchronization, and regulatory compliance oversight. Its structural boundaries, database queries, and view spaces cover the following operations:

- **User Identity Management and Role-Based Access Control Subsystem:**
System-wide identity management using clerk authentication utilities, role-based routing controls, user validation status flags, secure profile verification systems, and auditable action logs tracking administrative actions across the platform.
- **Analytics Dashboard and Epidemiological Surveillance Engine:** Rendering centralized tracking charts, regional heatmaps, and graphical data dashboards that analyze breeding efficiency percentages, municipal case counts, active dispatch queues, asset counts, and disease distribution trends across various local sectors.
- **Automated Regulatory Report Generation and Data Export Engine:** Compiling active database records to generate UNIP-compliant administrative PDF and Excel data reports, providing accurate, synchronized livestock summaries for formal submission to the Provincial Agriculture Office.

The automated scheduling, lifecycle tracking, and notification layers of the **BreedSmart** engine are strictly governed by hardcoded, breed-specific reproductive calculations. The algorithmic boundaries are confined to managing data changes, calculating target intervals, and firing automated reminders for specific reproductive milestones. The reheating or estrus return calculation module runs an automated background tracking window precisely 18 to 24 days following an active insemination entry to prompt users to watch for signs of fertilization failure. Alongside this, the gestation timeline calculator automatically computes expected delivery dates using specific constants mapped to local livestock breed traits. To manage long-term breeding health, the voluntary waiting period calculator tracks post-calving biological recovery windows before an individual animal asset is marked as eligible for subsequent breeding procedures.

The technology stack selected for building this distributed environment is strictly limited to specific languages, frameworks, and environments. The web application dashboard utilizes React, Vite, JavaScript, and TypeScript for the frontend administration view. The mobile application client utilizes React Native, Expo, Tailwind CSS, and NativeWind to build a responsive interface. The backend architecture implements Node.js and Express.js for the backend server API services. All data schemas, records, and system states are maintained through MongoDB Atlas as the cloud-hosted NoSQL repository for all core data structures.

1.7.2 Geographic, Biological, Hardware, and Infrastructure Constraints (The Delimitation)

To maintain an attainable and realistic software engineering scope for an undergraduate capstone study, specific parameters and design restrictions are established to separate the project from commercial, hardware-intensive, or nationwide configurations.

Geographical Boundaries and Local Government Unit Jurisdictional Delimitations

The geographical and local government boundaries strictly localize this study within the municipal borders of Oton, Iloilo. All field evaluation trials, data collection methods, user testing cycles, and synchronization simulations will exclusively involve farmers, field technicians, and administrative officials residing or working within the active barangays of this specific municipality. The system is not designed to support cross-border agricultural logistics, multi-municipal database clustering, or animal data ownership verification from adjacent towns or provinces.

Biological Classification Limits and Breed Core Concentration Boundaries

The biological core concentration optimizes the initial system core to prioritize data models, registries, and reproductive algorithms calibrated for the cattle industry or bovine livestock. This framework highlights commonly managed local breeds such as Brahman, Holstein, and local Bisaya crossbreeds. Features or data inputs for non-livestock animals, companion pets, or domestic poultry are explicitly excluded from this phase of the system. The system does not account for wild or exotic wildlife management records, equine asset registries, or non-bovine commercial livestock tracking.

Linguistic Boundaries and Regional Dialectical Localization Delimitations

The linguistic and dialect boundary establishes that while the software system incorporates localized natural language support via the Moowie AI assistant tool, the core system interface text, validation warnings, administrative forms, and error logs are delivered strictly in English and Tagalog. This study explicitly delimits its multi-language translation layer from regional structural variations of Hiligaynon (Ilonggo) or other specific Western Visayas sub-dialects spoken across different rural barangays in Oton. The system relies on standard urban

Hiligaynon datasets for conversational interactions and does not evaluate regional variations in dialect grammar or localized idioms.

Hardware Infrastructure Exclusions and IoT Architecture Delimitations

The hardware and IoT exclusions dictate that the **BreedSmart** platform is entirely software-reliant and depends completely on human-mediated data logging through web and mobile views. This study explicitly excludes the deployment, procurement, or integration of physical Internet of Things (IoT) sensors, wearable health monitors, GPS animal collars, automated biometric ear-tag readers, or automated drone surveillance networks. All identification practices rely entirely on reading alphanumeric tags manually clipped onto the animals and typing them into user forms.

Network Topology Constraints and Communication Infrastructure Delimitations

The network and infrastructure delimitations clarify that although the mobile client implements an offline caching architecture using AsyncStorage and TanStack Query to let field technicians save records in remote barangays with low signal, the application ultimately requires standard cellular networks (4G/5G) or Wi-Fi to sync local data queues with the cloud server. The development of alternative long-range network infrastructures, custom satellite interfaces, or localized mesh networks is outside the scope of this project.

Database Integration Boundaries and External API Interface Restrictions

The integration and API restrictions limit the database layer and reporting modules to function as a separate system for the Oton Municipal Agriculture Office. The architecture excludes direct API integrations with national livestock health platforms, national agricultural portals, or external

provincial databases. All report handovers to higher provincial units must be processed manually by downloading and printing the system's generated PDF or Excel data files.

1.8 Definition of Terms

The following terms are operationally and conceptually defined to provide a clearer understanding of the study and the proposed system:

Insemination Tracking

This is basically keeping a digital record of breeding activities — whether it's artificial or natural. The system logs breeding schedules, insemination dates, checks for pregnancy, and all the other reproductive tasks. BreedSmart handles all that.

Reheating Cycle

This is the period when a female cow goes back into heat after failing to get pregnant. The system figures this out automatically and keeps tabs on it. That way, technicians and farmers know when it's time to try breeding again.

Calf Drop

This means the exact moment or date a cow gives birth. BreedSmart logs these events so that reports are accurate and managing livestock gets a lot easier.

Breed Logic

This refers to the algorithms inside the system. They calculate breeding schedules, how long pregnancies last, and when cattle reach maturity — all based on different breeds like Brahman, Holstein, or Bisaya. It's what makes the automated tracking work.

Livestock Technician

These are government staff who handle breeding, insemination, vaccines, disease checks, and make sure data is correct out in the field, specifically in Oton. They're the main people using and managing BreedSmart.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Related Literature

Foreign Literature

Precision Livestock Farming (PLF) and Digital Transformation in Agriculture

Precision Livestock Farming (PLF) is one of the most significant developments in modern agriculture, focusing on the use of technology to monitor, manage, and optimize livestock production. According to the Food and Agriculture Organization (FAO, 2023), PLF integrates sensors, data analytics, and automated monitoring systems to improve animal health, reproduction efficiency, and farm productivity. Unlike traditional livestock management, which relies heavily on manual observation, PLF enables farmers to make data-driven decisions based on real-time information.

Wolfert et al. (2017) further explain that digital agriculture represents a shift from conventional farming practices to information-based systems where data plays a central role in decision-making. This transformation allows farmers to track individual animal performance, detect early signs of disease, and optimize breeding schedules. However, the study also highlights that the adoption of PLF is uneven, especially in developing countries where access to advanced technology is limited.

In relation to the proposed system, BreedSmart applies the core concept of PLF but in a simplified and accessible form. Instead of relying on expensive IoT devices such as sensors or wearable trackers, the system uses a web and mobile-based platform to manually and semi-automatically record livestock data. This approach ensures that even small-scale farmers in rural municipalities like Oton, Iloilo can benefit from digital livestock management without requiring high-cost infrastructure.

Digitalization of Livestock Breeding and Reproductive Management

Reproductive efficiency is a critical factor in livestock production, particularly in cattle farming. Hansen (2020) emphasizes that successful breeding management depends heavily on accurate detection of estrus cycles and timely insemination. Missing a single reproductive cycle can result in delayed calving, reduced milk production, and financial losses for farmers.

Similarly, Thatcher et al. (2019) found that modern reproductive technologies, including digital monitoring systems, significantly improve conception rates by ensuring that insemination is performed at the optimal time. These systems reduce human error by providing alerts and predictive analytics based on animal reproductive patterns.

Digital tools also help track pregnancy duration and expected calving dates, which are essential for proper livestock planning. Without such systems, farmers often rely on estimation and experience, which can lead to inaccurate breeding records and inefficient herd management.

This literature strongly supports the development of BreedSmart's automated breeding cycle computation. The system is designed to calculate reheat cycles, insemination intervals, and expected calf drop dates based on breed-specific reproductive data. This reduces manual estimation and improves accuracy in livestock management.

Global Adoption of Smart Farming Technologies

The World Bank (2022) reports that smart farming technologies are increasingly being adopted in developed agricultural economies due to their ability to improve productivity, reduce operational costs, and enhance food security. These technologies include digital record-keeping systems, automated livestock monitoring tools, and data-driven decision support systems.

Countries that implement smart farming systems benefit from improved traceability of livestock products, better disease control, and more efficient resource allocation. However, the report also highlights a major challenge: many developing regions still struggle to adopt these technologies due to limited infrastructure, lack of training, and financial constraints.

This situation is similar to rural areas in the Philippines, where livestock management is still largely paper-based. BreedSmart addresses this gap by offering a low-cost, accessible digital solution that does not require advanced hardware, making it suitable for local government units and small-scale farmers.

Digital Agriculture and Smallholder Farmers

According to the International Food Policy Research Institute (IFPRI, 2021), smallholder farmers often face barriers in adopting digital technologies due to limited technical knowledge and lack of access to digital infrastructure. Despite this, digital tools have been proven to significantly improve agricultural productivity when properly designed for local use.

IFPRI emphasizes that successful agricultural technologies must be simple, mobile-friendly, and context-specific to ensure adoption in rural communities. Complex systems that require advanced training or expensive equipment are less likely to succeed in developing regions.

This supports the design philosophy of BreedSmart, which prioritizes simplicity, mobile accessibility, and user-friendly interfaces tailored for farmers and livestock technicians in Oton,

Iloilo.

Mobile-Based Veterinary and Livestock Information Systems

Mobile technology has become an essential tool in improving agricultural and veterinary services, particularly in rural and hard-to-reach areas. According to Aker and Mbiti (2019), mobile-based systems significantly enhance information flow in agricultural communities by allowing farmers to report issues, receive guidance, and access services in real time. These systems reduce dependency on physical visits and manual reporting, which are often slow and inefficient.

In livestock management, mobile applications are increasingly used for disease reporting, vaccination tracking, and veterinary consultation. A study by Tadesse et al. (2022) found that mobile livestock platforms improved the speed of disease detection and response by enabling farmers to instantly report symptoms observed in their animals. This early reporting system helps prevent the spread of infectious diseases within herds, reducing economic losses.

In relation to BreedSmart, this literature supports the development of a mobile-based reporting and communication module. Farmers in Oton, Iloilo can use the system to quickly report animal diseases and request insemination services without needing to physically visit the Municipal Agriculture Office. This improves efficiency and ensures faster response from livestock technicians.

Artificial Intelligence and Predictive Livestock Management

Artificial Intelligence (AI) is increasingly being applied in livestock farming to predict animal health conditions, reproductive cycles, and productivity trends. According to Neethirajan (2020), AI-driven livestock systems analyze behavioral and physiological data to predict estrus cycles, disease risks, and optimal feeding schedules. These predictive systems allow farmers to make proactive decisions rather than reactive responses.

Similarly, Rutten et al. (2019) explain that data-driven livestock management improves decision-making accuracy by integrating historical records with real-time data inputs. Predictive analytics can identify patterns that are not easily observable through manual monitoring, especially in large herds.

However, the implementation of AI systems often requires high computational resources and advanced infrastructure, making them less accessible to small-scale farmers in developing regions.

BreedSmart addresses this limitation by applying simplified predictive logic instead of full AI systems. The system uses breed-based reproductive calculations to estimate key livestock events such as insemination timing, reheatings cycles, and expected calf drop dates. This provides similar benefits to predictive systems but in a more accessible and practical form for local agricultural use

Digital Disease Surveillance Systems in Livestock Farming

Disease surveillance is a critical component of livestock management, especially in preventing outbreaks that can affect entire communities. According to the World Organisation for Animal

Health (WOAH, 2023), early detection and reporting systems are essential in controlling livestock diseases and ensuring food security.

Research by Del Rio Vilas et al. (2020) highlights that digital surveillance systems improve disease reporting accuracy and reduce delays in government response. Traditional paper-based systems often suffer from missing data, delayed submission, and inconsistent reporting formats, which weaken disease control efforts.

Digital surveillance platforms, on the other hand, allow centralized data collection and real-time monitoring of livestock health conditions. These systems enable agricultural authorities to quickly identify disease clusters and implement containment measures.

In relation to the proposed study, BreedSmart incorporates a simplified disease reporting module that allows farmers to report livestock health issues directly to livestock technicians. This ensures faster communication and more accurate disease monitoring at the municipal level.

Smart Agriculture and Data-Driven Decision Making

Smart agriculture refers to the integration of digital technologies, data analytics, and automation in farming practices. According to Rose and Chilvers (2018), smart agriculture improves decision-making by providing farmers with accurate and timely data regarding crop and livestock conditions.

Data-driven systems help reduce uncertainty in farming operations by replacing guesswork with measurable indicators. In livestock farming, this includes monitoring reproductive cycles, tracking health conditions, and analyzing productivity trends.

However, the adoption of smart agriculture technologies is still limited in rural communities due to cost, complexity, and lack of technical training. Many farmers continue to rely on traditional methods, which are less efficient and more prone to errors.

BreedSmart aligns with smart agriculture principles by providing a simplified digital platform that focuses on usability and accessibility. Instead of complex AI-driven systems, it uses structured data inputs and automated calculations that are easy for farmers and technicians to understand and use.

Summary of Foreign Literature

The reviewed foreign literature consistently highlights the importance of digital transformation in livestock management. Technologies such as mobile applications, predictive analytics, AI systems, and digital surveillance have significantly improved livestock productivity, disease control, and reproductive efficiency in developed agricultural systems.

However, most of these technologies require advanced infrastructure and technical expertise, which limits their application in rural and small-scale farming communities. This gap highlights the need for simplified, accessible systems such as BreedSmart that can deliver similar benefits without requiring complex hardware or high-level technical knowledge.

Local Literature

Digital Transformation in Philippine Agriculture

The Department of Agriculture (DA, 2022) emphasizes that digital transformation is a key strategy in modernizing Philippine agriculture. The agency recognizes that traditional farming practices, while still widely used, are no longer sufficient to meet the growing demands for food security, efficiency, and accurate agricultural reporting. As a response, the DA continues to

promote the use of digital tools such as mobile applications, centralized databases, and farm monitoring systems to improve agricultural productivity.

However, despite these initiatives, adoption remains slow in many rural areas. According to DA reports, one of the main barriers is the lack of access to reliable internet connectivity and the limited digital literacy of farmers. Many livestock keepers still rely on handwritten logs, verbal communication, and manual tracking of breeding cycles. This situation often leads to inconsistent records and delayed reporting to local government offices.

In relation to the present study, BreedSmart addresses these issues by providing a simple and mobile-accessible system designed specifically for local farmers and livestock technicians in the Municipality of Oton, Iloilo. The system is designed to be user-friendly, ensuring that even users with minimal technical background can navigate and use it effectively.

Importance of Livestock Industry in the Philippine Economy

According to the Philippine Statistics Authority (PSA, 2024), the livestock industry plays a significant role in the country's agricultural economy, contributing billions of pesos annually in production value. Livestock such as cattle, swine, and poultry are essential sources of income, food supply, and rural livelihood.

Despite its importance, the livestock sector continues to face challenges related to inefficient record-keeping, disease management, and lack of centralized data systems. PSA also reports that inaccurate or delayed livestock data can affect national planning, disease response strategies, and agricultural policy development.

This highlights the need for improved data management systems at the local level. BreedSmart supports this requirement by providing a centralized database for livestock records, helping

ensure that data submitted by the Municipal Agriculture Office of Oton is accurate, complete, and updated in real time.

Technology Adoption Challenges in Rural Communities

The International Livestock Research Institute (ILRI, 2021) explains that smallholder farmers in developing countries often face barriers in adopting modern agricultural technologies. These barriers include limited financial resources, lack of technical training, and insufficient access to digital infrastructure.

ILRI further notes that technology designed for rural communities must prioritize simplicity, affordability, and usability. Complex systems that require advanced hardware or technical expertise are less likely to be adopted by small-scale farmers.

In the Philippine context, many rural municipalities still rely on manual systems for livestock monitoring. This creates inefficiencies in data collection, breeding management, and veterinary coordination. BreedSmart addresses this gap by providing a lightweight, web and mobile-based system that does not require expensive equipment or advanced technical knowledge.

Digital Agriculture and Local Government Units (LGUs)

Local Government Units (LGUs) play a crucial role in implementing agricultural programs and managing livestock data at the municipal level. According to studies on LGU digitalization initiatives in the Philippines, many municipalities still struggle with integrating digital systems into their agricultural services due to limited budgets and infrastructure constraints.

As a result, many agricultural offices continue to use paper-based records, which are prone to loss, damage, and data inconsistency. This affects the accuracy of reports submitted to provincial and national agricultural agencies.

BreedSmart is designed to support LGU operations by providing a centralized system for livestock data management. It enables livestock technicians to record breeding activities, monitor animal health, and generate reports more efficiently, reducing the burden of manual documentation.

Livestock Management Practices in Rural Areas

In rural farming communities, livestock management is often based on traditional knowledge passed down through generations. While this method is valuable, it lacks precision in terms of reproductive tracking, disease monitoring, and data analysis.

Studies in rural agricultural systems in the Philippines show that farmers often depend on memory or handwritten logs to track insemination dates and calving schedules. This increases the risk of errors, missed breeding cycles, and inaccurate reporting.

The absence of digital tools also makes it difficult for livestock technicians to maintain updated and synchronized records across different barangays or farms. BreedSmart addresses these limitations by introducing automated tracking of breeding cycles, including reheating periods and expected calf drop dates, ensuring more accurate and reliable livestock management.

Summary of Local Literature

The reviewed local literature highlights the ongoing challenges in Philippine livestock management, particularly in rural municipalities. While national agencies such as the Department of Agriculture and PSA promote digital transformation, implementation at the local level remains limited due to infrastructure and accessibility issues.

Smallholder farmers and LGUs continue to rely on manual systems, which are inefficient and prone to errors. This creates a clear need for a localized, simple, and accessible digital solution.

BreedSmart addresses this gap by providing a centralized livestock management system designed specifically for municipal-level use. It improves data accuracy, enhances communication between stakeholders, and supports the modernization of livestock management practices in the Municipality of Oton, Iloilo.

2.2 Related Studies

Mobile-Based Livestock Disease Reporting Systems

A study conducted by Gizaw et al. (2023) in Ethiopia examined the effectiveness of mobile-based livestock disease reporting systems in rural farming communities. The study found that using mobile applications significantly improved the speed and accuracy of disease reporting compared to traditional paper-based methods. Field workers were able to submit real-time reports, which allowed veterinary authorities to respond faster and prevent the spread of infectious diseases.

The study also highlighted that delays in manual reporting often result in uncontrolled outbreaks, which can severely affect livestock populations and farmers' income. By contrast, mobile systems reduced communication gaps between farmers and government veterinary services.

In relation to BreedSmart, this study supports the development of a mobile-based disease reporting module that allows farmers in Oton, Iloilo to instantly report livestock health concerns to technicians. This ensures faster response time and better disease control at the municipal level.

Digital Reproductive Monitoring and Conception Efficiency

Research conducted in Kazakhstan (2026) on digital reproductive monitoring systems found that automated estrus detection significantly increased conception rates in cattle. The system provided timely alerts for optimal insemination periods, reducing missed breeding opportunities.

The study concluded that reproductive efficiency improved because farmers no longer relied solely on manual observation, which is often inaccurate or delayed. Instead, digital systems provided predictive indicators that guided breeding decisions.

This supports BreedSmart's automated breeding cycle feature, which calculates reheat cycles, insemination schedules, and expected calf drop dates based on breed-specific reproductive data.

Mobile Veterinary Support Systems

A study in Nigeria (2025) evaluated the impact of mobile veterinary support applications in rural farming areas. The findings revealed that mobile platforms improved access to veterinary assistance, especially in regions where livestock technicians are limited in number.

Farmers reported faster communication with veterinary personnel and improved disease management outcomes due to real-time messaging and reporting features. The study emphasized that mobile systems help bridge the gap between farmers and agricultural experts.

This is highly relevant to BreedSmart, which aims to improve communication between farmers and livestock technicians through a centralized mobile and web platform.

IoT-Based Livestock Monitoring Systems

Research in India and China on IoT-based livestock monitoring systems shows that sensor-based tracking improves livestock productivity and reduces losses caused by disease and poor management. According to Kumar et al. (2022), IoT systems can monitor temperature, movement, feeding behavior, and reproductive status in real time.

However, the study also notes that IoT systems are expensive and require advanced infrastructure, making them less accessible to small-scale farmers in developing countries.

BreedSmart addresses this limitation by offering a software-based alternative that does not require physical sensors but still provides structured monitoring through automated calculations and user input.

Local Studies

Artificial Insemination Success in Central Luzon

A study conducted by the Department of Agriculture Region III (2021) found that proper estrus detection and accurate record-keeping significantly improve artificial insemination success rates in cattle. Farmers who used structured or digital records achieved higher conception rates compared to those relying on manual logs.

The study emphasized that synchronization between farmers and livestock technicians is essential for improving reproductive outcomes.

This directly supports BreedSmart's centralized livestock database and automated breeding tracking system, which ensures that both farmers and technicians have access to updated and consistent data.

IoT and Smart Farming Studies in Laguna

A study conducted in Laguna (2022) showed that IoT-based livestock systems reduced feed waste by up to 30% and improved farm efficiency through real-time monitoring. The research highlighted the importance of digital data collection in optimizing livestock production.

Although IoT systems provided high accuracy, the study also noted that implementation costs remain a major limitation for rural farmers.

This supports the need for a simplified system like BreedSmart, which provides similar benefits through software-based monitoring instead of expensive hardware.

Technological Impact on Livestock Record Management in Leyte

Dadulla (2017) conducted a study in Leyte focusing on the transition from manual livestock record-keeping to digital systems. The findings revealed that digital systems significantly improved data accuracy, reduced record loss, and enhanced reporting efficiency to agricultural offices.

The study concluded that paper-based systems are no longer sufficient for modern livestock management due to their vulnerability to damage and human error.

This supports BreedSmart's core objective of replacing manual logs with a centralized digital system for livestock monitoring and reporting

Mobile Agriculture Systems in the Philippines

Local research on mobile agriculture systems in the Philippines shows that mobile-based platforms improve communication between farmers and agricultural offices. According to studies conducted by local agricultural development programs, farmers are more likely to adopt systems that are simple, accessible, and available on mobile devices.

However, challenges such as low digital literacy and inconsistent internet connectivity remain barriers in rural areas.

This reinforces the importance of designing BreedSmart as a lightweight, mobile-friendly system that can function even in low-resource environments.

Summary of Related Studies

The reviewed foreign and local studies consistently show that digital systems significantly improve livestock management, particularly in disease reporting, reproductive monitoring, and communication between stakeholders.

Foreign studies emphasize advanced technologies such as IoT, AI, and mobile veterinary systems, while local studies highlight the challenges of adopting such technologies in rural Philippine communities.

Despite these advancements, a gap remains in the availability of simple, affordable, and municipality-level livestock management systems. BreedSmart addresses this gap by

providing a centralized, mobile-based solution designed specifically for local government use in Oton, Iloilo.

2.3 Synthesis

The reviewed literature and studies presented in this chapter collectively show that livestock management has significantly evolved through the integration of digital technologies such as mobile applications, precision livestock farming systems, artificial intelligence, and IoT-based monitoring tools. Both foreign and local studies consistently emphasize that digital transformation improves livestock productivity, enhances disease control, and increases the accuracy of reproductive and health-related records.

Foreign literature highlights advanced innovations such as Precision Livestock Farming (PLF), AI-based reproductive prediction systems, and IoT-enabled monitoring devices. These technologies allow real-time tracking of animal health, automated detection of reproductive cycles, and improved decision-making based on data analytics. Studies from FAO (2023), World Bank (2022), and related researchers confirm that these systems contribute to higher efficiency and better livestock management outcomes in modern agricultural settings.

However, despite their effectiveness, most of these technologies require high-cost infrastructure, technical expertise, and continuous internet connectivity. This makes them less suitable for small-scale farmers and rural communities in developing countries, particularly in the Philippines.

Local literature and studies further reveal that Philippine livestock management still largely relies on manual and paper-based systems. The Department of Agriculture (DA, 2022) and Philippine Statistics Authority (PSA, 2024) both indicate that while the livestock sector plays a major role in

the national economy, it continues to face challenges in data accuracy, reporting delays, and lack of centralized digital systems. Smallholder farmers in rural municipalities such as Oton, Iloilo often depend on handwritten records and verbal communication, which increases the risk of data loss, errors, and inefficiencies in breeding management.

Related studies also show that mobile-based systems, even in simplified forms, significantly improve livestock monitoring, disease reporting, and reproductive management. Studies from Ethiopia, Nigeria, Kazakhstan, and the Philippines demonstrate that digital platforms enhance communication between farmers and agricultural authorities, reduce response time in disease outbreaks, and improve breeding success rates. However, many of these systems are either too complex, dependent on IoT hardware, or not specifically designed for local government agricultural operations.

Based on these findings, there is a clear gap between advanced livestock management technologies and the actual needs of rural farming communities in the Philippines. Existing solutions are either too advanced for local adoption or too limited in scope for municipal-level agricultural management.

This gap is what the proposed system, **BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System**, aims to address. The system combines the essential features of modern livestock management—such as automated breeding cycle computation, digital record-keeping, disease reporting, and real-time communication—into a simplified and accessible platform.

Unlike IoT-heavy systems, BreedSmart focuses on software-based automation that does not require expensive hardware. It is designed specifically for livestock technicians and farmers in the Municipality of Oton, Iloilo, ensuring usability even for users with limited technical

background. Through breed-specific logic for Brahman, Holstein, and Bisaya cattle, the system provides accurate calculations for reheating cycles, insemination schedules, pregnancy timelines, and expected calf drop dates.

In conclusion, the synthesis of related literature and studies confirms the necessity of a localized, user-friendly, and centralized livestock management system. BreedSmart bridges the gap between advanced agricultural technologies and practical rural implementation by providing a digital solution that is both functional and accessible for local government agricultural operations.

CHAPTER 3

METHODOLOGY

3.1 System Development Life Cycle (SDLC)

The study adopts a **developmental research design** using the **Agile Software Development Model** combined with the **System Development Life Cycle (SDLC)** approach. This methodology is suitable for the iterative development of BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System, allowing continuous improvement based on user feedback from farmers and livestock technicians in the Municipality of Oton, Iloilo.

The SDLC framework ensures that the system is developed in a structured and systematic process, from planning up to deployment. Each phase focuses on improving system quality, usability, and functionality to meet the actual needs of the end users.

Planning Phase

The planning phase involves identifying the overall scope of the project and understanding the livestock management situation in Oton, Iloilo. Preliminary site visits and informal interviews with livestock technicians and farmers are conducted to gather initial insights.

During this phase, the researchers also study common issues in cattle management such as manual record-keeping, difficulty in tracking breeding cycles, and delays in reporting livestock health conditions. These observations help define the objectives of the system and establish its relevance to real-world agricultural problems

Requirements Analysis Phase

In this phase, the system requirements are gathered and analyzed based on the needs of the target users, which include livestock technicians, farmers, and the Municipal Agriculture Office.

Key requirements identified include:

- Digital recording of livestock information
- Automated tracking of insemination, reheating, and calf drop dates
- Easy communication between farmers and technicians
- Centralized storage of livestock records
- Generation of accurate reports for administrative use

This phase ensures that the system is aligned with actual user needs rather than assumptions, improving its effectiveness and usability.

System Design Phase

The system design phase focuses on creating the blueprint of the BreedSmart system. This includes designing the user interface (UI), user experience (UX), system architecture, and database structure.

The UI design prioritizes simplicity and accessibility, especially for users with limited technical experience. The system is designed to be mobile-responsive to support farmers who primarily use smartphones.

The database design uses MongoDB to store livestock records, breeding history, vaccination data, and user accounts. The system architecture is structured to separate frontend and backend processes for better scalability and maintainability.

Development Phase

The development phase involves the actual coding and implementation of the system. The frontend is developed using **React.js**, while the backend is built using **Express.js** with **Node.js**, and **MongoDB** is used as the database management system.

During this phase, key system modules are developed, including:

- Farmer request module (insemination and disease reporting)
- Technician management module (breeding and health monitoring)
- Automated breeding cycle calculator based on cattle breed logic
- Notification system for reminders and updates

- Administrative dashboard for the Municipal Agriculture Office

This phase follows Agile principles, allowing iterative improvements based on continuous evaluation.

Testing Phase

The testing phase ensures that the system functions correctly and meets user requirements.

Different levels of testing are performed, including:

- **Unit Testing** – testing individual system components
- **System Testing** – testing the complete integrated system
- **User Acceptance Testing (UAT)** – involving actual users such as livestock technicians and selected farmers

This phase helps identify errors, bugs, and usability issues before full deployment. Feedback from users is also collected to improve system performance and interface design.

Finalization and Deployment Phase

In the final phase, the system is prepared for deployment and presentation. All documentation is completed, including user manuals and technical documentation.

The system is finalized based on feedback gathered during testing. After validation, BreedSmart is prepared for implementation in a controlled environment within the Municipality of Oton, Iloilo.

This phase ensures that the system is stable, functional, and ready for real-world use in livestock management operations.

3.2 Project Gantt Chart (Weekly Estimation)

The timeline spans 21 weeks, from December 10, 2025, to May 2, 2026.

Phase	Task	Duration	Timeline
I. Planning	Preliminary Research, Site Visits, and Proposal Preparation.	Weeks 1–13	Dec 10 – Mar 10
II. Requirement Analysis	Interviews, Questionnaires & Data Gathering	Weeks 14–15	Mar 11 - Mar 24
III. Design	System Design: Blueprinting UI/UX and database schemas	Weeks 16–27	Mar 25 – Jun 11
IV. Development	Coding & Database Setup	Weeks 27 – 27	Jun 11 – Jun 18
V. Testing	Bug Fixing & Integration	Week 28	Jun 18 – Jun 19

VI. Submission	Document submission	Week 28	Jun 20
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3.3 Comparative Benchmark: BreedSmart System Features

To contextualize the proposed Livestock Management and Veterinary Service System for Oton, Iloilo, this study examines BreedSmart (coupled with the Moowie mobile client), a contemporary agricultural and veterinary services platform. BreedSmart demonstrates how localized systems integrate advanced features such as real-time livestock tracking, offline-first sync engines, veterinary gestation calculators, multilingual AI assistant support, and administrative dashboards. By analyzing its architecture, the study identifies both strengths and limitations that inform the design of the localized Iloilo-focused system.

Complete Feature List

Frontend (User-Facing Modules)

- **Livestock Registry:** Profile creation, digital ear-tag registration, breed classification (Beef Cattle, Dairy Cattle, Carabao, Swine, Goat), dynamic age tracking, status labels, and image attachments.
- **Service Request Portal:** Quick-actions for requesting Artificial Insemination (AI) or veterinary health checks, symptoms reporting, and image attachments.
- **Breeding & Gestation Calendar:** Interactive heat-cycle calendars, voluntary waiting period (VWP) tracking, diagnostic check windows, and expected calving calculators.
- **Technician Dashboard:** Task actions, walk-in registration sheets, queue logs, and dynamic expected diagnostic dates visualization.

- Reports & Analytics: Graph charts displaying breeding efficiency, municipal case counts, and automated UNIP-compliant PDF exports for record-keeping.
- User Accounts: Clerk authentication, OTP verification, secure password reset, profile editing, and role-based permissions (Admin, Technician, Farmer).
- AI Chat Support (Moowie AI): Multilingual guidance (Hiligaynon, Tagalog, English), livestock wellness recommendations, diagnostic help, system guidance, and conversational memory.
- Admin Features: Role assignment, user verification, system-wide analytics, data exports, log management, and system settings.
- Landing Page: General information, service overviews, public reports, and download links for the mobile client.

Backend (System Services)

- Activity & Sync Tracking: Queues offline actions using client-side storage, with automatic resolution and server-side synchronization upon reconnection.
- Security & Protection: Encrypted sessions, validation schema guards, API rate limiting, and role-based security checks.
- Notifications: Real-time web socket alerts, automated mobile pushes, status update reminders.
- Image Uploads: Cloud storage integration for animal verification and health report photos.
- Queue System: Background task handler managing sync retries, notifications, and scheduled calving updates.
- Location Features: Barangay-level coordinates caching, regional heatmaps, route optimization coordinates.
- Soft Delete System: Cascading soft deletes using timestamp indicators, preserving historical breeding ledgers for auditing.
- API Versioning: Supports multiple versions for feature expansion.
- Multi-Environment Support: Local development, staging, and production databases.

AI Chatbot Capabilities (Moowie AI)

- Language detection and localized Hiligaynon, Tagalog, and English greetings.
- Symptom diagnostics and first-aid recommendations.
- Breeding cycle guidance (step-by-step).
- Vet-care recommendations.
- Interactive help menus and FAQs.
- Follow-up question handling with context memory.

Security Summary

- JWT tokens, role-based routing middleware, Clerk Identity protection, API rate limiting, auditable action logs, and validation schemas.

Technology Stack

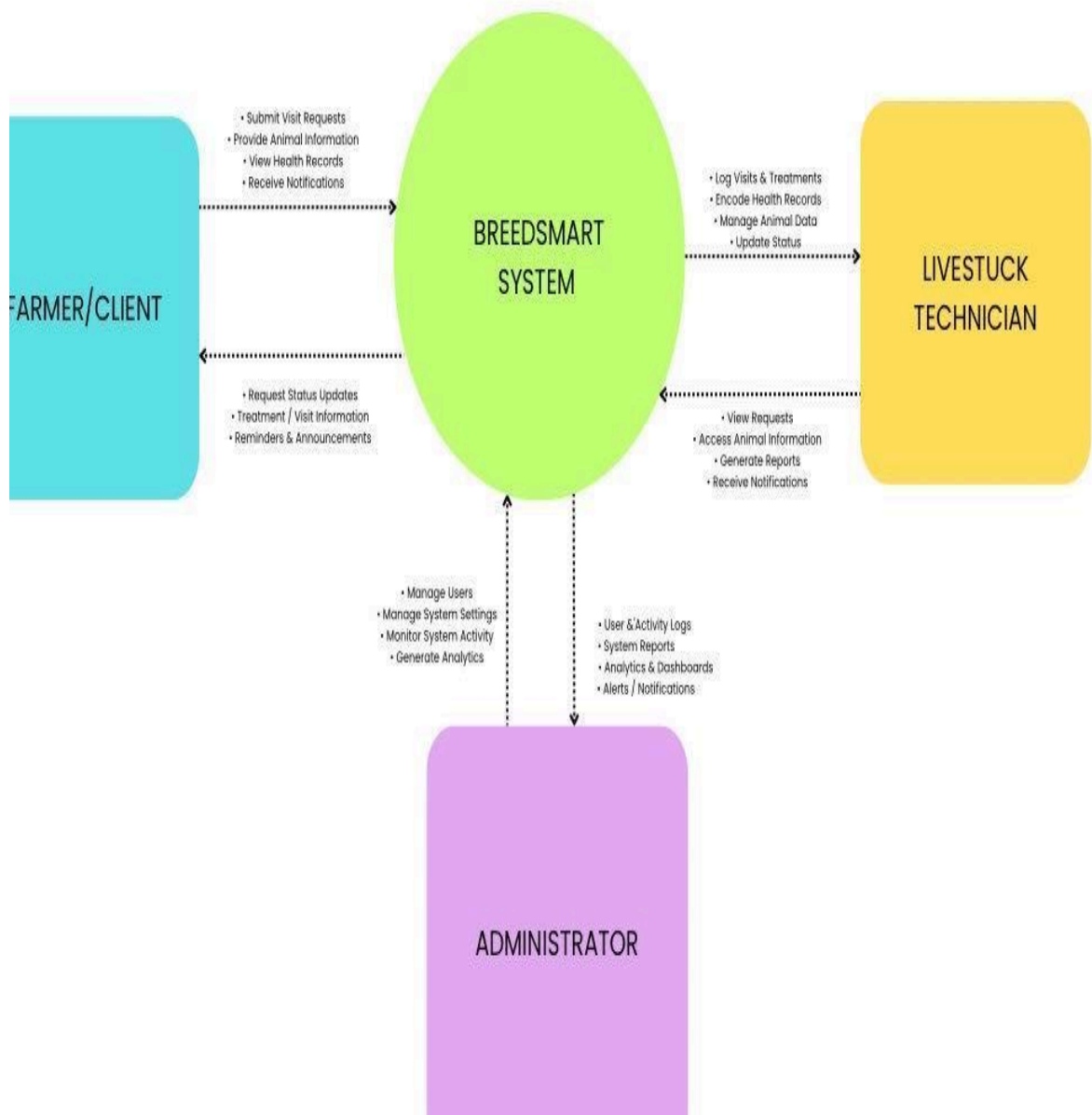
- Frontend: React, Vite, React Native, Expo, Tailwind CSS, NativeWind, DaisyUI, Leaflet maps.
- Backend: Node.js, Express, Clerk.
- AI: Google Gemini API.
- Real-time: Socket.io.
- Database & Caching: MongoDB, Mongoose, AsyncStorage, TanStack Query.
- Queue: Inngest.
- Storage: Cloudinary.

3.4 Requirement Modeling

3.4.1 Context Diagram

The **Context Diagram** serves as a high-level representation of the entire BreedSmart ecosystem, distilling the platform into a singular process to clearly define the external entities and their primary interactions with the system.

- **Farmer / Client:** Responsible for submitting personal profiles, livestock specifications, and service requests for veterinary or field visits; receives automated registration confirmations and synchronized visit schedules in return.
- **Livestock Technician:** Acts on real-time alerts for service requests and performs direct digital entry of health treatments, vaccinations, and insemination records to eliminate manual logbook dependency.
- **Agriculture Officer / Admin:** Oversees health status summaries and total livestock populations while utilizing the system to generate comprehensive analytical reports for municipal and provincial administrative oversight.



3.5 Data Structures

The BreedSmart system utilizes **MongoDB Atlas** as its centralized, cloud-based NoSQL repository for secure data storage and management. This approach ensures high reliability and data redundancy for critical agricultural information, moving away from vulnerable paper-based logs. The data structure is organized into several key collections designed to support the multi-tiered platform for Farmers, Technicians, and Administrators. The primary collections are:

- **Users:** Stores profile information for all system stakeholders, including the distinct roles of Administrator, Livestock Technician, and Farmer. This segregation of user profiles is essential for managing access levels and defining user-specific actions within the mobile and web interfaces.
- **Animals:** This collection is fundamental for the system's core function, tracking vital livestock data such as species, specific breed (e.g., Brahman, Holstein, Bisaya), and current health status. It serves as the foundation for automating breeding cycle calculations based on specific **Breed Logic**.
- **HealthRecords:** A detailed log documenting all health-related interventions. This includes records of specific treatments, comprehensive vaccination history, and mortality rates. This collection ensures the Oton Agriculture Office can generate accurate reports for submission to the Provincial level.

```
"User" : {  
  
  "_id" : ObjectId  
  
  "clerkId" : "String" // unique, sparse  
  
  "name" : "String" // required  
  
  "email" : "String"  
  
  "phoneNumber" : "String"  
  
  "role" : "admin|technician|farmer" // default: farmer  
  
  "status" : "active|on-site|on-leave" // default: active  
  
  "isVerified" : "Boolean" // default: false  
  
  "pushToken" : "String" // Expo push token  
  
  "address" : { AddressSchema }  
  
  "createdAt" : "Date" // timestamps  
  
  "updatedAt" : "Date" // timestamps  
  
}
```

```
"Animal" : {  
  
  "_id" : ObjectId  
  
  "farmerId" : ObjectId // → User  
  
  "motherId" : ObjectId // → Animal  
  
  "animalId" : "String" // unique tag  
  
  "earTag" : "String"  
  
  "species" : "Beef|Dairy|Carabao|Goat..."  
  
  "breed" : "String" // required  
  
  "gender" : "Male|Female" // default: Female  
  
  "reproductiveStatus" : "Normal|InHeat|Pregnant..."  
  
  "birthDate" : "Date"  
  
  "parity" : "Number" // birth count, default 0  
  
  "sireDetails" : { breed, code }  
  
  "bcsHistory" : [{ score, recordedAt }] // body condition scores  
  
  "activityLogs" : [{ event, date, desc }] // embedded array  
  
  "isVerified" : "Boolean" // default: false  
  
}
```

```
"Insemination" : {  
  
  "_id" : ObjectId  
  
  "farmerId" : ObjectId // → User  
  
  "animalId" : ObjectId // → Animal  
  
  "technicianId" : ObjectId // → User  
  
  "approvedBy" : ObjectId // → User  
  
  "pregnancyId" : ObjectId // → Pregnancy  
  
  "inseminationDate" : "Date"  
  
  "preferredDate" : "Date"  
  
  "scheduledDate" : "Date"  
  
  "sireBreed" : "String"  
  
  "sireCode" : "String"  
  
  "estrus" : "Natural|Sync|Induced"  
  
  "status" : "pending|approved|done..." // default: pending  
  
  "outcome" : "Pending|Pregnant|Failed.." // default: Pending  
  
  "attemptNumber" : "Number" // default: 1  
  
  "technicianNote" : "String"  
  
}
```

```
"Pregnancy" : {  
  
  "_id" : ObjectId  
  
  "animalId" : ObjectId // → Animal  
  
  "farmerId" : ObjectId // → User  
  
  "inseminationId" : ObjectId // → Insemination  
  
  "pregnancyDiagnosis" : { date, result }  
  
  "targetCalvingDate" : "Date" // expected calving  
  
  "technicianNote" : "String"  
  
}
```

```
"AIRequest" : {  
  
  "_id" : ObjectId  
  
  "farmerId" : ObjectId // → User  
  
  "animalId" : ObjectId // → Animal  
  
  "handledBy" : ObjectId // → User  
  
  "preferredDate" : "Date"  
  
  "scheduledDate" : "Date"  
  
  "status" : "pending|approved|rejected|done" // default: pending  
  
  "imageUrl" : "String"  
  
  "comment" : "String"  
  
  "technicianNote" : "String"  
  
}
```

```
"HealthRequest" : {  
  
  "_id" : ObjectId  
  
  "farmerId" : ObjectId // → User  
  
  "animalId" : ObjectId // → Animal  
  
  "handledBy" : ObjectId // → User  
  
  "requestType" : "disease|medicine|checkup|injury"  
  
  "symptoms" : "String" // required  
  
  "urgency" : "low|medium|high" // default: medium  
  
  "status" : "pending|approved|resolved..."  
  
  "preferredDate" : "Date"  
  
  "scheduledDate" : "Date"  
  
  "diagnosis" : "String"  
  
  "treatment" : "String"  
  
  "advice" : "String"  
  
  "technicianNote" : "String"  
  
}
```

hk

```
"Task" : {  
  
  "_id" : ObjectId  
  
  "technicianId" : ObjectId // → User  
  
  "farmerId" : ObjectId // → User  
  
  "animalIds" : ObjectId // → [Animal]  
  
  "taskType" : "AI|PD|CD|Vaccination|Other"  
  
  "category" : "Urgent|Routine|Follow-up" // required  
  
  "priority" : "Number" // 1=High 2=Med 3=Low  
  
  "status" : "Pending|In Progress|Done"  
  
  "notes" : "String" // required  
  
}
```

```
"FieldNote" : {  
  
  "_id" : ObjectId  
  
  "technicianId" : ObjectId // → User  
  
  "farmerId" : ObjectId // → User  
  
  "title" : "String" // required  
  
  "description" : "String"  
  
  "farmerName" : "String" // denormalized snapshot  
  
  "latitude" : "String"  
  
  "longitude" : "String"  
  
  "locationName" : "String" // default: Oton, Iloilo  
  
  "imageUrl" : "String"  
  
}
```

3.6 System Architecture

The system follows a **client-server architecture**, a model strategically selected to address the systemic inefficiencies identified in the municipality's current manual workflows. By centralizing data processing and storage while distributing access via various interfaces, this architecture eliminates data fragmentation and ensures that all stakeholders interact with a single, synchronized source of truth. This centralized control is vital for maintaining the integrity of livestock records, while the modular nature of the architecture provides the scalability necessary to accommodate an increasing number of users and animals without compromising system performance.

- **Expanded Client Layer:** This layer comprises the primary interfaces for user interaction, bifurcated into a web-based administrative portal and a cross-platform mobile client. The web portal is optimized for high-density data management at the Agriculture Office, providing administrators with tools for comprehensive reporting and oversight. Conversely, the mobile client, developed using React Native, is tailored for the specific needs of farmers and field technicians. To bridge the digital literacy gap, the interface prioritizes intuitive navigation and accessibility. Crucially, the mobile layer implements a robust offline data caching engine using AsyncStorage, allowing technicians to record field observations in remote barangays with poor connectivity, with data automatically synchronizing once a network connection is re-established.
- **Expanded Server Layer:** Acting as the "brain" of BreedSmart, the server layer—built on Node.js and Express.js—governs all system logic and data transit. It is responsible for processing complex API requests from the client layer and enforcing strict security protocols, including user authentication and role-based authorization via Clerk. Furthermore, the server executes the core business logic of the application, such as the breed-specific reproductive

calculations that automate the tracking of reheating cycles and calving windows. By centralizing these computations, the system ensures that every breeding milestone is calculated with mathematical precision, removing the risk of human error associated with manual estimation.

- **Expanded Database Layer:** The system utilizes MongoDB Atlas, a NoSQL database, to manage its centralized repository. The choice of a NoSQL approach is particularly beneficial for agricultural data, which often requires schema flexibility to accommodate evolving metrics. As new health parameters or breeding traits are identified for tracking, the document-based structure of MongoDB allows for seamless data model updates without the need for intensive database overhauls. This ensures that the system remains adaptable to the changing requirements of the livestock sector in Oton, providing a resilient foundation for long-term records including insemination histories, vaccination logs, and maternal lineages.

Data Synchronization and Security

Communication between these layers is governed by strict API security standards and real-time synchronization protocols. All data transit is encrypted via HTTPS to maintain the confidentiality of sensitive agricultural records. To ensure data integrity, the system utilizes validation schemas that audit incoming data before it is committed to the database. Real-time synchronization—powered by Socket.io—ensures that as soon as a technician logs a treatment in the field, the information is immediately reflected on the administrative dashboard, creating a seamless information pipeline between the farm and the municipal office.

3.7 Tools and Technologies Used

The development of BreedSmart utilizes modern technologies to ensure system responsiveness, scalability, and maintainability.

- **Frontend:** React, JavaScript, TypeScript, Tailwind CSS, NativeWind, Vite, and Expo are used to develop responsive and user-friendly web and mobile interfaces.
- **Backend:** Express.js and Node.js are used for server-side processing, API development, and system logic implementation.
- **Database:** MongoDB serves as the database management system for storing livestock records, breeding information, and user data.
- **Development Tools:** Visual Studio Code is used for coding and debugging, MongoDB Compass for database management, and GitHub for version control and collaboration.

These technologies were selected for their compatibility with Agile development practices, open-source availability, and reliability in modern application development.

3.8 Data Gathering Techniques

The researchers utilized several data gathering techniques to collect relevant information needed for the development of BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System.

- **Surveys/Questionnaires:** Structured questionnaires were distributed to selected farmers and livestock personnel to assess current challenges related to manual record-keeping, communication delays, livestock monitoring, and accessibility of agricultural services within different barangays in Oton, Iloilo.

- **Interviews:** Direct interviews and consultations were conducted with the livestock technician and selected farmers to gather detailed information about breeding cycles, insemination procedures, livestock health monitoring, and existing reporting practices.
- **Observations:** The researchers conducted physical observations and reviews of existing hardbound logbooks, folders, and manual livestock records to identify issues such as data degradation, record inconsistencies, and inefficiencies in the current documentation process.

3.9 Testing Procedures

The researchers conducted several testing procedures to ensure the functionality, reliability, and overall performance of the BreedSmart system. These testing methods were performed to identify system errors, validate features, and confirm that the system meets the requirements of the intended users.

- **Unit Testing** – Unit testing was conducted to evaluate individual components and modules of the system separately. Each function, including livestock record management, breeding cycle calculations, notification features, and authentication processes, was tested to ensure proper operation and accurate outputs.
- **System Testing** – System testing was performed to evaluate the integrated functionality of the entire BreedSmart system. This testing ensured that all modules work together correctly, including communication between the frontend, backend, and database. The researchers also tested system responsiveness, data synchronization, and overall workflow performance.
- **User Acceptance Testing (UAT)** – User Acceptance Testing was conducted with selected farmers, livestock technicians, and intended users to determine whether the system meets their operational needs and expectations. Feedback gathered during this phase was used to identify

usability issues, improve user experience, and validate the practicality of the system in a real-world agricultural environment.

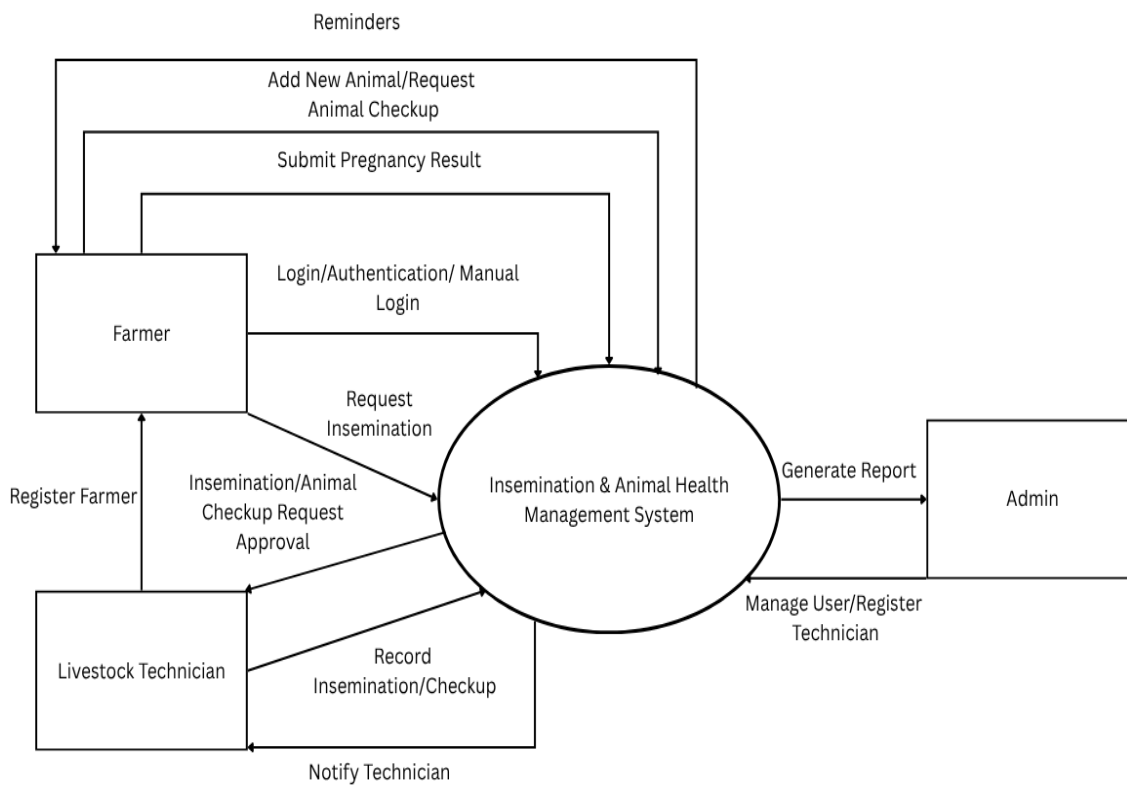
3.10 Evaluation Criteria

The BreedSmart system will be evaluated based on selected software quality characteristics from the ISO 25010 standard to ensure that the system is reliable, effective, and suitable for livestock management operations in the Municipality of Oton, Iloilo.

- **Functionality** – Evaluates the ability of the system to perform its intended functions accurately and efficiently. This includes livestock record management, breeding cycle monitoring, notification services, and report generation.
- **Usability** – Measures how easy and user-friendly the system is for farmers, livestock technicians, and administrators. This includes interface design, accessibility, navigation, and overall user experience.
- **Accuracy** – Assesses the correctness and reliability of automated breeding calculations, rearing schedules, calf drop estimations, and livestock monitoring records generated by the system.
- **Security** – Evaluates the system's ability to protect user accounts, livestock records, and sensitive information from unauthorized access and data loss through authentication and secure data handling.
- **Efficiency** – Measures the responsiveness and performance of the system during data processing, record retrieval, notification handling, and overall system operations across web and mobile platforms.

3.11 Supporting Diagrams

Data Flow Diagram (DFD – Level 0)



Farmer Registration & Animal Managemen

Farmer / Client → BreedSmart → Database

- Farmer submits personal profile, farm location, and animal details

- System processes the registration and links the animals to the farmer's profile
- Data is stored securely in the database and a confirmation is returned to the Farmer

Health Record & Treatment Logging

Technician → BreedSmart → Database

- System associates the treatment data with the correct farm and animal records
- Database updates the animal's health status and saves the historical health record
- Technician records health treatments, vaccinations, Insemination and diagnoses for specific animals

Visit Request & Scheduling

Farmer / Client → BreedSmart → Technician

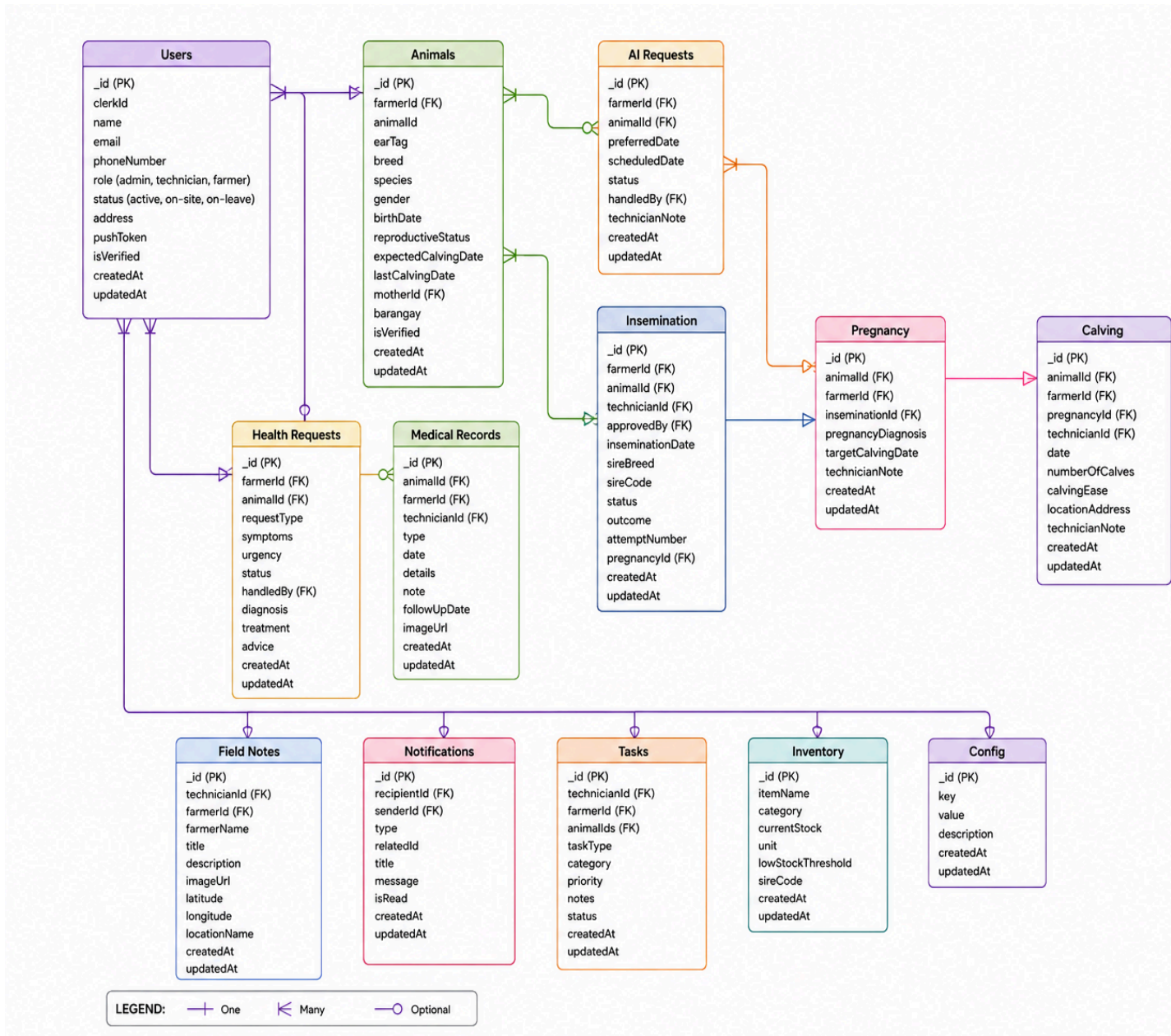
- Farmer submits a request for a veterinary or farm visit through the system
- System processes the request and alerts the assigned Technician
- Technician reviews the request, schedules the visit, and the system sends the scheduled date back to the Farmer.

Analytics and Reporting

Agriculture Officer / Admin → Agriculture Office System → Database

- Officer or Admin requests a summary of health statuses or total livestock population
- System queries the database for aggregated animal and health data
- System compiles the findings, generates analytical reports, and displays them to the Officer/Admin

Entity Relationship Diagram (ERD)



Entities

Users

- _id (PK)

- clerkId
- name
- email
- phoneNumber
- role
- status
- address
- pushToken
- isVerified
- createdAt
- updatedAt

Animals

- _id (PK)
- farmerId (FK)
- animalId
- earTag
- breed
- species
- gender
- birthDate
- reproductiveStatus
- expectedCalvingDate
- barangay
- motherId (FK)

AI Requests

- `_id` (PK)
- `farmerId` (FK)
- `animalId` (FK)
- `preferredDate`
- `scheduledDate`
- `status`
- `handledBy` (FK)

Insemination

- `_id` (PK)
- `animalId` (FK)
- `technicianId` (FK)
- `inseminationDate`
- `sireBreed`
- `status`
- `pregnancyId` (FK)

Pregnancy

- `_id` (PK)
- `animalId` (FK)
- `inseminationId` (FK)
- `targetCalvingDate`
- `pregnancyDiagnosis`

Calving

- `_id` (PK)
- `animalId` (FK)
- `pregnancyId` (FK)
- `technicianId` (FK)
- `date`
- `numberOfCalves`

Health Requests

- `_id` (PK)
- `farmerId` (FK)
- `animalId` (FK)
- `requestType`
- `symptoms`
- `urgency`
- `status`

Medical Records

- `_id` (PK)
- `animalId` (FK)
- `technicianId` (FK)
- `type`
- `date`
- **`details`**

Notifications

- `_id` (PK)
- `recipientId` (FK)
- `senderId` (FK)
- `title`
- `message`
- `isRead`

Tasks

- `_id` (PK)
- `technicianId` (FK)
- `farmerId` (FK)
- `taskType`
- `priority`
- `status`

Inventory

- `_id` (PK)
- `itemName`
- `category`
- `currentStock`
- `unit`

Config

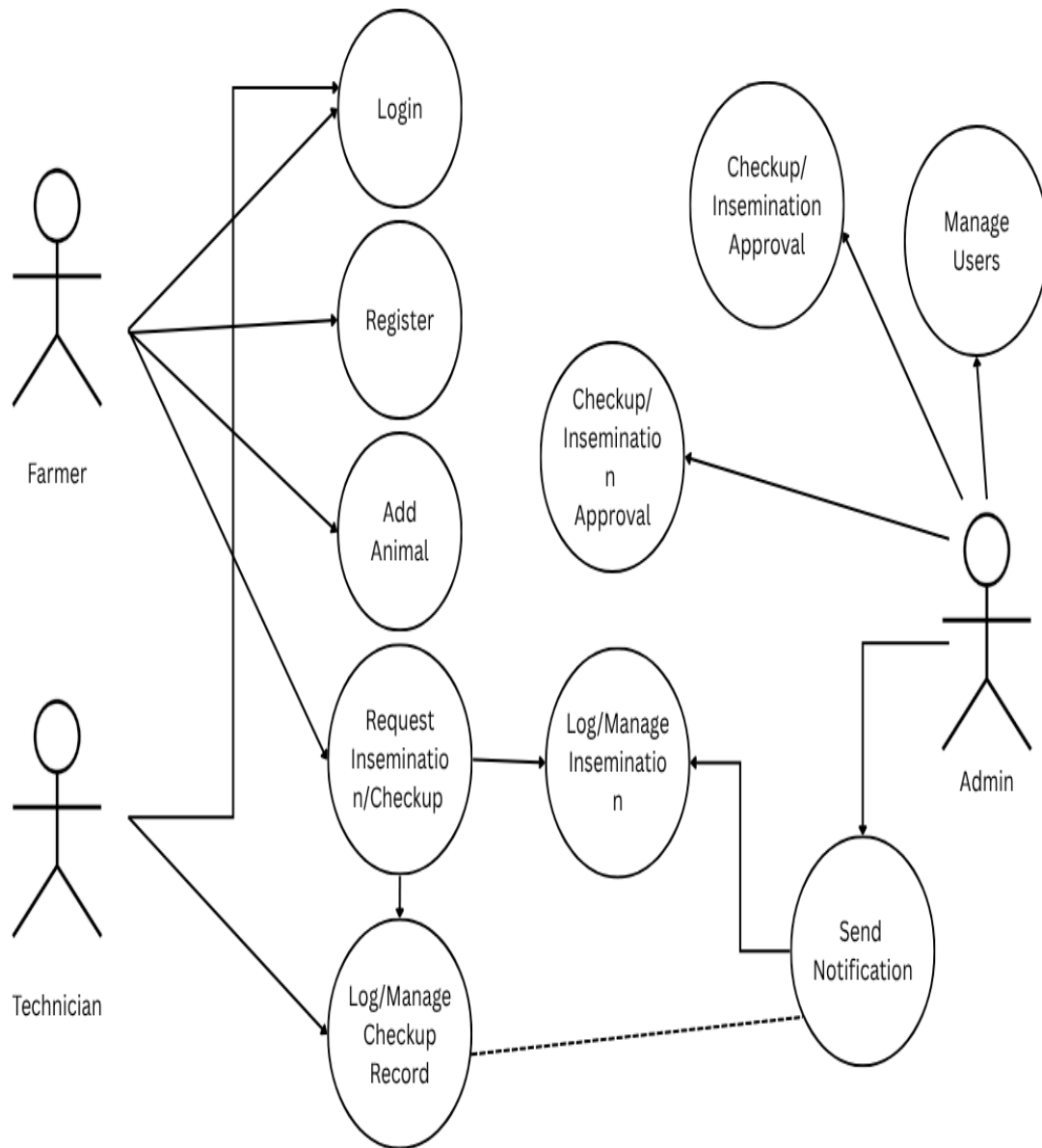
- `_id` (PK)
- `key`
- `value`

- description

Relationships

- User to Animal: One farmer can own multiple animals.
- User to AI Request: One farmer can submit multiple insemination requests.
- User to Health Request: One farmer can submit multiple health-related requests.
- Animal to Insemination: One animal can have multiple insemination records.
- Insemination to Pregnancy: One insemination process may result in one pregnancy record.
- Pregnancy to Calving: One pregnancy record can result in one calving event.
- Animal to Medical Record: One animal can contain multiple medical records.
- Technician to Task: One technician can manage multiple livestock-related tasks.
- User to Notification: Users can send and receive multiple notifications.
- Animal to Animal: One animal may have a maternal relationship through motherId.

Use Case Diagram



Actors

1. Farmer / Client

- Farmer → Register Account → Log In
- Farmer → Update Account Profile
- Farmer → Add and Manage Animal Records
- Farmer → Request Insemination Services
- Farmer → Request Veterinary or Health Checkup
- Farmer → View Animal Health History
- Farmer → Receive Notifications and Schedule Reminders
- Farmer → Monitor Pregnancy and Calving Records

2. Livestock Technician

- Technician → Log In
- Technician → Review Insemination Requests
- Technician → Approve and Schedule Farm Visits
- Technician → Record Insemination Procedures
- Technician → Record Treatments and Vaccinations
- Technician → Manage Animal Health Records
- Technician → Update Pregnancy and Calving Information
- Technician → Send Notifications and Follow-Up Reminders

3. Administrator

- Administrator → Log In
- Administrator → Manage User Accounts and Roles

- Administrator → Verify Farmers and Technicians
- Administrator → Monitor Livestock Records
- Administrator → Manage Insemination and Health Requests
- Administrator → Generate Analytical and Administrative Reports
- Administrator → View System Activities and Statistics
- Administrator → Send Announcements and Notifications

3.12.1 Interface Diagram (Wireframe Design)

Website Design

Figure 3.1: Web Wireframe Interface Design — Central Administrative Dashboard View

The visual layout presented details the **Central Administrative Dashboard View** of the BreedSmart web portal, engineered specifically for deployment within the Municipal Agriculture Office of Oton, Iloilo. This interface serves as the primary data aggregation window for municipal administrators, providing a structured environment for rapid information retrieval and decision-making. The architectural framework features a persistent vertical navigation menu, a workspace identification header detailing session states, and a multi-grid analytical canvas optimized for high-density data visibility.

The interactive telemetry cards display real-time livestock population tallies, including Total Users, Registered Farmers, Active Technicians, and Verified Pregnancies, derived from the system's underlying MongoDB Atlas data states. Beneath these informational metrics, the application layout partitions active processes into large-scale graphical visualizations, incorporating a "System Throughput" area chart and a "District Density" data matrix to monitor livestock population trends and health case distribution across various local sectors. The bottom grid introduces a "Dispatch Queue" activity ticker alongside a "Registry Log" canvas, allowing office personnel to audit pending veterinary requests before deploying field technicians.

Programmatically, initializing this view triggers a series of aggregated server-side requests through Node.js and Express.js API routers to scan the **Users**, **Animals**, and **Tasks** collections. The client-side dashboard UI leverages React components to display data fields, ensuring seamless rendering and rapid page responsiveness. This design approach satisfies the rigorous

performance, efficiency, and usability characteristics mandated by the project's selected ISO 25010 software quality evaluation framework.

Figure 3.2: Web Wireframe Interface Design — Field Personnel and Technician Fleet Roster

The interface mockup presented illustrates the **Technician Fleet Roster** layout of the administrative web suite, which serves as the central utility for managing local government field personnel. The structural layout maintains the vertical system navigation toolbar for consistent usability, while the main content area displays a grid of interactive employee profile container blocks. This view is designed to provide municipal agriculture administrators with a clear visual summary of all field personnel assigned to different agricultural zones within the municipality.

Each card template incorporates a placeholder icon, a label for the employee's name, credential data rows, and action triggers. The dashboard backend connects directly with the **Users** database collection, running filtered server queries where the user property definition is evaluated as **role: "technician"**. Each technician block dynamically lists individual status indicators such as *active*, *on-site*, or *on-leave*, ensuring real-time staff transparency for administrative dispatchers.

When a desk administrator selects a specific account profile card, the interface loads a detailed overlay sub-view. This view lists historical field notes, performance metrics, and individual schedule logs tracked under the technician's unique ID. This design effectively eliminates paper-based shift logs, which are prone to damage and human error, providing a more reliable way for the Municipal Agriculture Office to coordinate and monitor farm visit activities.

Figure 3.3: Web Wireframe Interface Design — Global Animal Registry Database

The wireframe layout presented depicts the **Global Animal Registry Database** panel, serving as the primary data interface for managing livestock inventory records within Oton. The visual

framework contains an input text widget for search parameters, a dropdown menu selector to filter records by municipal zones or barangays, and an interactive data grid that arranges animal record blocks into organized columns. Each entry container features a circular asset avatar placeholder, an identity field displaying unique system identification codes, and data lines for specific breed details.

The application UI populates this view by executing a server-side query on the centralized **Animals** collection in MongoDB Atlas. The view aggregates information by linking individual cattle profiles to their owner's record using the **farmerId** property. It displays essential parameters such as the unique alphanumeric **earTag** identifier, breed profiles (such as Brahman, Holstein, or Bisaya crossbreeds), biological gender, and maternal lineage tracking links (**motherId**).

Administrators can utilize the integrated filter tools to scan the global database in real time, reducing the lookup delays associated with manual paper-based logs. Toggling row actions triggers an administrative update modal, allowing office staff to verify ear-tag updates, review field notes, check veterinary histories, and audit data inputs before generating official compliance reports for the Provincial Agriculture Office.

Figure 3.4: Web Wireframe Interface Design — Artificial Insemination Ledger Matrix

The visual blueprint depicted outlines the **Artificial Insemination Ledger Matrix**, a specialized data view integrated into the administrative portal to support breeding management. The wireframe implements a dense layout to incorporate comprehensive process parameters into a single view, organizing records into columns for Date, Status, Animal Details, Owner, Technician, Sire Information, and Action triggers.

The application populates this screen by running lookups across the **Insemination**, **Animals**, and **Users** collections. The dashboard tracks the status of field operations, updating status labels to reflect values such as *pending*, *approved*, or *done* based on real-time entries made by field technicians. It also logs critical outcome parameters such as *pregnant* or *failed* to evaluate the success rate of local artificial breeding initiatives.

Administrators can utilize the context-menu action triggers to resolve scheduling conflicts, assign backup field workers to pending tasks, and view uploaded health images. This centralized data pipeline prevents record inconsistencies, balances field worker workloads, and provides a reliable database for analyzing breeding success trends across the municipality.

Figure 3.5: Web Wireframe Interface Design — System User Profiles and Access Management

The interface blueprint presented outlines the **System User Profiles** panel, which handles identity management and system-wide security controls for the web platform. The layout features a centralized data view containing a search textbox, an access level filter, a barangay zone selection list, and a tabular record layout. Data columns are organized into clear fields for User Identity, Roles & Status, Contact Information, and Actions.

The administrative backend coordinates this interface by connecting with the **Users** database collection, displaying active profiles for Administrators, Livestock Technicians, and Farmers. The view shows verification checkboxes and user status states, such as *active* or *on-site*, ensuring robust access control across the platform.

Administrators can utilize row commands to verify newly registered accounts, modify security permission levels, update farm contact information, and audit system activity logs. This centralized user management protects sensitive agricultural records from unauthorized access and data loss, fulfilling the security standards required by the ISO 25010 evaluation matrix.

Figure 3.6: Web Wireframe Interface Design — Municipal Census and Analytical Report Hub

The mockup presented details the **Municipal Census and Analytical Report Hub**, serving as the primary interface for generating official statistics and data exports. The layout features control areas for selecting date ranges, along with download buttons to export data as UNIP-compliant PDFs or Excel files. The main view area uses a grid layout to display multiple data charts, including a "Barangay Activity" chart and a "Status Mix" visualization panel.

When an administrator triggers a report request, the backend server processes data across the **Animals**, **HealthRecords**, and **Insemination** collections. The platform aggregates data points to generate statistics on total livestock populations, breeding success rates, and local disease case distributions across different sectors.

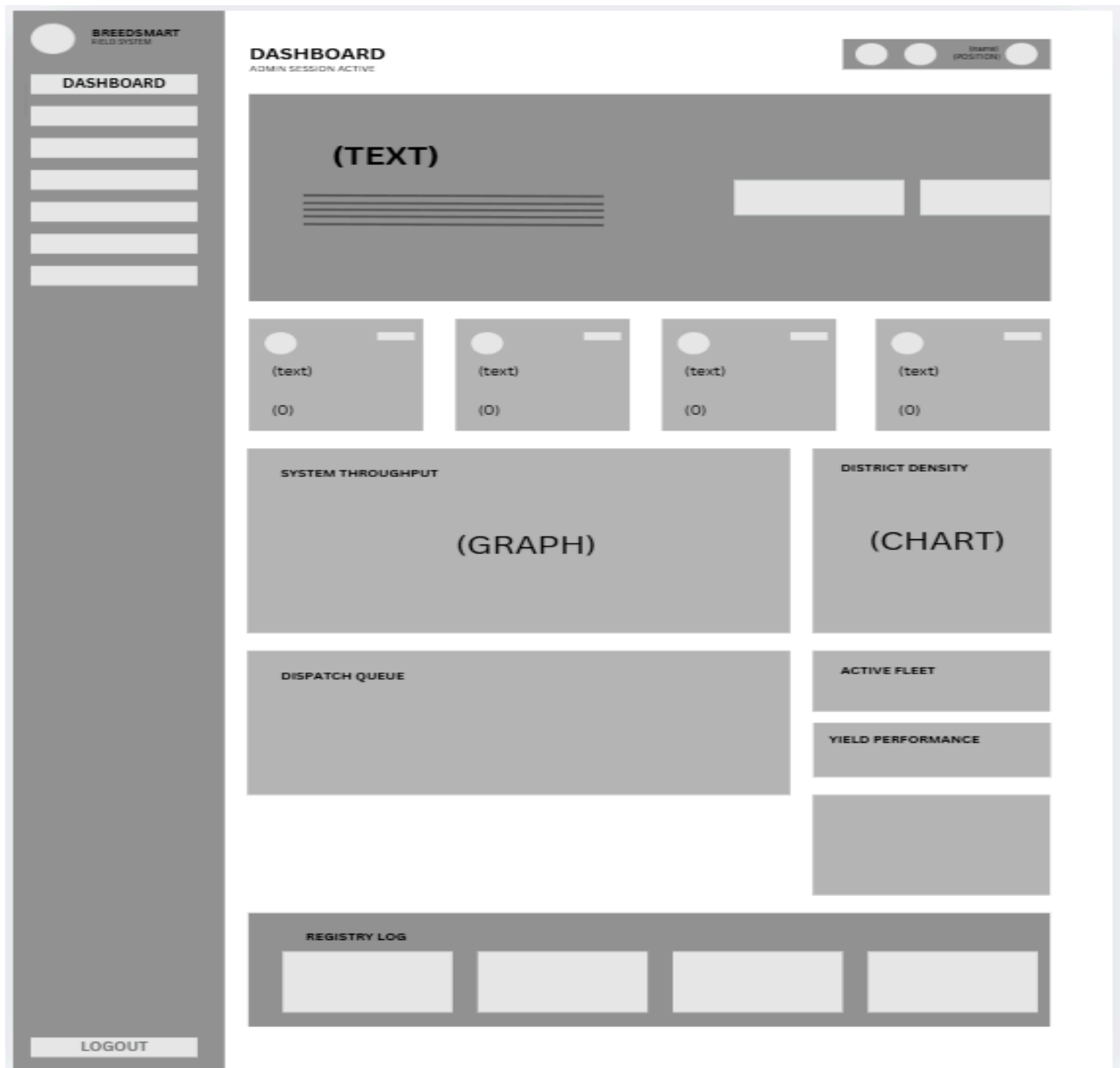
Activating the export buttons triggers backend scripts that compile active records into formal administrative data files. This automated report generation allows the Municipal Agriculture Office to submit accurate, synchronized summaries to the Provincial Agriculture Office quickly, effectively replacing the inefficient process of compiling paper records manually.

Figure 3.7: Web Wireframe Interface Design — Central System Control and Profile Settings

The wireframe layout presented illustrates the **Central System Control and Profile Settings** interface, which manages configuration settings for the web workspace. The layout features a split design consisting of a left-aligned sub-navigation block, a main system control container, and an individual profile details area. The profile column features a user icon placeholder, text strings for employee details, and data modification fields.

The application coordinates this view by mapping values to the **Config** database collection. Authorized staff can utilize the panel to adjust default system rules, such as updating the

notification alert windows used by the automated breeding tracker. Individual users can utilize the profile form to update login credentials, manage communication preferences, and verify security configurations, keeping the system stable and secure while allowing for easy administrative tuning.







BREEDSMART
FIELD SYSTEM

Placeholder bar

Placeholder bar

LIVESTOCK

Placeholder bar

Placeholder bar

Placeholder bar

Placeholder bar

Placeholder bar

LOGOUT

LIVESTOCK

ADMIN SESSION ACTIVE

ANIMAL REGISTRY

GLOBAL LIVESTOCK DATABASE

Team:
(POSITION)

SEARCH

ALL ZONES



(#0)

Placeholder bar

Placeholder bar



(#0)

Placeholder bar

Placeholder bar



(#0)

Placeholder bar

Placeholder bar



(#0)

Placeholder bar

Placeholder bar



(#0)

Placeholder bar

Placeholder bar



(#0)

Placeholder bar

Placeholder bar



BREEDSMART
FIELD SYSTEM

Placeholder bar

Placeholder bar

Placeholder bar

INSEMINATIONS

Placeholder bar

Placeholder bar

Placeholder bar

LOGOUT

LIVESTOCK

ADMIN SESSION ACTIVE



(name)
(POSITION)



DATE STATUS ANIMAL DETAILS OWNER TECHNICIAN SIRE INFO ACTION

(DATA)



BREEDSMART
FIELD SYSTEM

-
-
-
-
-
- USERS
-
-

LOGOUT

USERS

ADMIN SESSION ACTIVE

(name)
(POSITION)

SYSTEM USER

ADMIN SESSION ACTIVE

SEARCHALL ROLESALL BARANGAY

USER IDENTITY	ROLES & STATUS	CONTACT INFO	ACTIONS
(DATA)			



BREEDSMART
FIELD SYSTEM

-
-
-
-
-
-
-
- REPORTS
-

REPORTS

LOGOUT

REPORTS

ADMIN SESSION ACTIVE



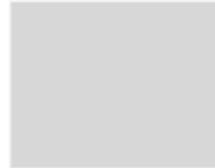
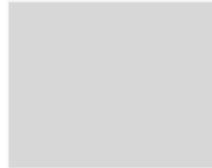
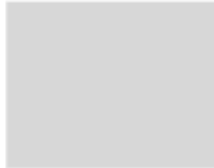
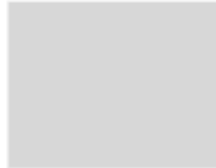
MUNICIPAL CENSUS HUB

Official reporting and analytics for the Municipality of Otton.

DATE

PDF

EXCEL



BARANGAY ACTIVITY

(CHART)

STATUS MIX

(CHART)



BREEDSMART
FIELD SYSTEM

-
-
-
-
-
-
-
-

SETTINGS

LOGOUT

SETTINGS

ADMIN SESSION ACTIVE

(name)
(POSITION)

SYSTEM CONTROL PANEL

-
-
-
-
-



(NAME)

(DETAILS)

Mobile Design

Figure 3.1: Mobile Wireframe Interface Design — Secure Identity Gateway and Portal Login Screen

The wireframe layout presented depicts the **Secure Identity Gateway** mobile view, serving as the primary authentication portal for both farmers and livestock technicians. The visual template features a clean layout containing a branding element, a "Sign in with Google" button, credential input fields, a "Forgot password?" link, and a navigation trigger to register a new account profile.

The mobile application handles authentication by integrating with secure identity validation systems. When a user submits credentials, the app forwards the request to the Node.js API server to verify session permissions against encrypted records stored in the **Users** MongoDB collection.

Depending on the authorized role, the application automatically routes the user to either the Farmer panel or the Technician field dashboard. This gatekeeper design ensures that the centralized database is protected from unauthorized modifications and that users only access authorized features and records.

Figure 3.2: Mobile Wireframe Interface Design — Central Dashboard Overview and Quick Actions View

The visual layout presented illustrates the **Central Dashboard Overview**, which serves as the primary landing space upon mobile authentication. The top header displays profile metadata, user roles, and live timestamp markers. Beneath this, a search bar allows users to scan for specific records, while an informational summary card shows core system tallies. The bottom grid provides quick-action buttons for streamlined feature navigation.

The client application populates the overview panel by querying user-specific endpoints. For livestock technicians, the summary card displays live municipal counts for Total Users, Registered Farmers, Active Technicians, and Monitored Pregnancies. Quick-action buttons route users to functional areas such as All Users, All Animals, Create User, or Records Log. The client interface leverages modern mobile responsive design principles, ensuring smooth transitions even on lower-end devices, thereby satisfying usability and efficiency metrics.

Figure 3.3: Mobile Wireframe Interface Design — User Directory and Profile Registry List Screen

The mockup presented details the **User Directory and Profile Registry List** screen, designed to manage user records within the mobile client application. The layout features a header containing an access role filter block and a text search bar, with the main display region consisting of multi-line container blocks designed to display detailed user summary records cleanly.

The client app updates this list by querying the **Users** database collection, categorizing views into All, Farmer, Technician, and Admin buckets to facilitate rapid contact location. Profile cards display essential details such as verification flags, contact strings, and active field assignments. Livestock technicians can utilize these cards to review farmer requests, update status flags, and check background histories, effectively replacing traditional paper lists for field activity coordination.

Figure 3.4: Mobile Wireframe Interface Design — Contextual Admin Actions Bottom Sheet

The wireframe view presented illustrates the **Contextual Admin Actions Bottom Sheet** overlay, designed to display contextual tools based on user roles. The layout features an overlay panel that glides up from the bottom menu bar, revealing an action grid with options such as Create User, All Animals, Records Log, and All Users. The mobile client handles this overlay

using state-driven UI layers, providing fast access to secondary workflows for authorized technicians or managers. Tapping any option triggers a clean interface transition to the selected workflow, ensuring features are organized cleanly on smartphone screens and improving data management efficiency.

Figure 3.5: Mobile Wireframe Interface Design — Global Animals Directory Directory Screen

The interface blueprint presented details the **Global Animals Directory** screen, which acts as the primary tool for searching and updating livestock records in the field. The layout includes a top search widget optimized for entering alphanumeric tag values, followed by a vertical stack of multi-line data blocks displaying individual animal histories.

The mobile app populates this listing by scanning the **Animals** database collection, displaying tracking parameters such as unique ear-tag identifiers, breed profiles (Brahman, Holstein, or Bisaya), gender, and active reproductive status labels. Field personnel can select these records to access full medical histories, review previous treatment logs, and update health details during farm visits, ensuring informed decision-making on-site and replacing the use of loose notes.

Figure 3.6: Mobile Wireframe Interface Design — Reproductive Ledger Logs and Tabbed Matrix Screen

The visual mockup presented details the **Reproductive Ledger Logs** view, which tracks breeding milestones and reproductive cycles. The header contains button triggers to sort records into categories: Inseminations, Pregnancies, and Calvings. The mobile app handles this screen by loading or clearing cached records as users switch between tabs. Selecting "Inseminations" lists operational flows from the **Insemination** collection, while "Pregnancies" queries the automated breed logic engine to display pregnancy check windows and delivery estimates. This tabbed layout enables field workers to track time-sensitive breeding events easily, minimizing data errors and ensuring compliance with breed-specific timelines.

Figure 3.7: Mobile Wireframe Interface Design — Task Management and Operations Dispatch Monitor

The wireframe layout presented details the **Task Management** mobile monitor, acting as the primary interface for managing field schedules. The screen uses a card layout to organize field tasks into a vertical checklist, incorporating priority banners, job descriptions, location details, and status checkboxes. The application coordinates this workflow by reading data from the **Tasks** database collection, automatically assigning tasks based on approved service requests or vaccination deadlines. Technicians can utilize these lists to open maps, log field treatments, and sync results back to the office dashboard, ensuring field staff and office dispatchers remain synchronized and improving agricultural service delivery.

BreedSmart Farmer Mobile Client Application Layer

The farmer-centric interface of the **BreedSmart** mobile platform is engineered as a highly synchronized, decentralized gateway aimed at mitigating communication latency between cattle producers and the Municipal Agriculture Office of Oton, Iloilo. Functioning as the client-side tier within a distributed client-server architecture, the application utilizes an offline-first caching framework supported by local storage and asynchronous state queries. The user experience is anchored by a persistent bottom navigation bar featuring five primary nodes: HOME, Animals, a dynamic central Action Trigger (+), Records, and Profile. This navigation logic facilitates seamless transit through specialized, role-scoped functional modules designed to maintain high data fidelity during field operations.

Core Functional Modules & Interface Layers

Central Workspace Home and Live Telemetry Dashboard

- **My Farm Status Counter:** Upon successful authentication, the primary sub-view initializes a real-time telemetry card titled "My Farm Status." This dashboard executes database-driven queries to aggregate herd data, partitioning metrics into three live status blocks—Waiting Result, Pregnant, and Calving—enabling raisers to monitor active reproductive cycles at a glance.
- **Quick Actions Floating Matrix:** Located beneath the herd telemetry cards, a row of interactive quick-action buttons provides immediate navigational paths to essential agricultural services, including direct triggers for Request AI, Request Vet, Add Cow, and Ask Moowie.
- **Moowie AI Conversational Widget:** The base of the home dashboard incorporates a dedicated conversation window featuring the "Moowie AI" assistant. This interface provides context-aware natural language support, offering farmers immediate access to multilingual guidance, troubleshooting protocols, and step-by-step veterinary recommendations.

AI Outcome Checker and Time-Sensitive Alert Feed

- **Interactive Lifecycle Interventions:** When reproductive timelines reach critical verification milestones, the workspace dynamically displays the AI Outcome Checker. This module addresses field estimation vulnerabilities by prompting the farmer to provide explicit confirmation via triggers: Yes, Pregnant! or No, Re-heat.
- **Upcoming Visits Monitor:** Features an active tracking strip for scheduled technical farm visits, outlining the service type, target window, assigned technician, and a prominent Cancel trigger for real-time schedule modifications.

- **Heat & Breeding Alerts Feed:** Contained within a visual alert zone, this vertical stack displays chronological alerts calibrated by breed-specific logic, featuring scrolling counters for upcoming calving dates and pending pregnancy check windows.

Digitized Service Request & Disease Triage Intake Portals

- **Request AI Service Form:** This digital portal replaces paper schedules and verbal reporting. The interface features an auto-populated "Your Information" card, a dropdown Select Animal component, selectors for Preferred Visit Date and Time, a camera container to Attach Photo for remote validation, and a free-text box for technical Notes.
- **Report Animal Issue Form:** Structured to ensure consistent usability, this diagnostic portal includes a Moowie Support contextual banner. Below user metadata, it provides a Request Type dropdown and a three-tier, color-coded urgency toggle—Low, Medium, and Urgent—enabling rapid dispatch for severe symptoms.

Asset Inventory Management & Digital Registration Engine

- **My Animals Global Directory:** This screen provides a comprehensive catalog of registered cattle. The header includes an asynchronous search box for ear-tag lookups, while the main scroll canvas displays a vertical stack of livestock profiles with unique codes, breed details (e.g., Brahman, Holstein, or Bisaya), and navigation to veterinary charts.
- **Add to My Farm Registration Panel:** Accessed via the expansion button, this layout facilitates real-time herd expansion. It includes a camera modal for verification photos, an alphanumeric Ear Tag field with an automated generator, and standardized menus for Species, Breed, Color, Gender, and Birth Date inputs.

State-Driven Quick Utilities & Contextual Bottom Sheets

- **Quick Actions Bottom Sheet:** Activating the navigation plus button (+) triggers a sliding modal layer. The overlay darkens the background to focus on a six-button grid listing shortcuts: Request AI, Report Issue, My Requests, Add Animal, Disease Map, and Ask Moowie, providing a flat hierarchy that eliminates nested menus.

Comprehensive Service Tracker & Production Analytics Center

- **Chronological Service Hub:** Acts as an administrative pipeline for requests. A filter bar organizes tickets into All, Pending, In-Progress, and Requests. Active rows utilize badge tags to indicate task states (e.g., Done or Resolved), detailing attempt counts, sire information, and links to animal histories.
- **Reports Center Production Dashboard:** This hub is divided into Breeding Cycles and Activity Feed tabs. The Breeding Cycles panel displays automated alerts with color-coded countdowns to minimize human error. The Activity Feed provides a metadata counter row above a continuous ledger of calving records and validated health changes.
-



BreedSmart



Sign in with Google

OR

Email or Username

Password

[Forgot password?](#)

Sign In

Don't have an account? [Register](#)

(Date)



(Name)
(Position)



Greetings
(text)

(0)
MISSIONS

(0)
AI CYCLE

(0)
CLINICAL

Quick Actions

Record AI

Health Log

Add Client

Add Animal

Pregnancy

Calf Drop

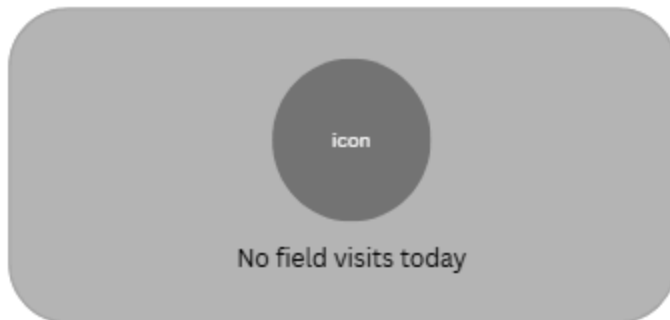
(Date)



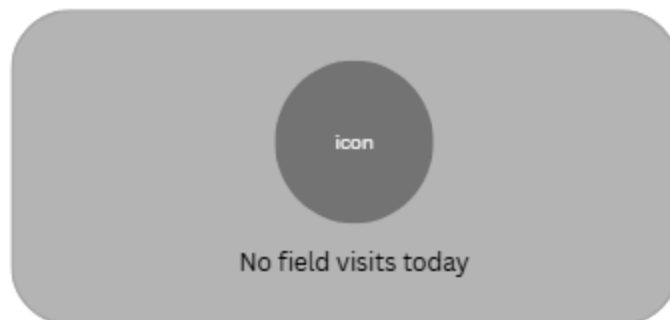
(Name)
(Position)



Today's Route



Farmer Requests



HOME



USERS



ANIMALS



LEDGER

← Record AI

FARMER SELECTION

[REGISTER NEW](#)

icon

Select Farmers...



A.I PROCEDURE DETAILS

SERVICE MODE

Complete

Schedule

SIRE BREED

Select Sire Breed...



SIRE CODE/BULL ID

SIRE-9P3T

MISSION DATE

TIME

(DATE)

(TIME)

ESTRUS CYCLE / TYPE

Natural

Synchronized

Induced

ADDITIONAL NOTES

Any other details about the procedures...

← Health log

Existing Record

Manual Entry

ASSIGN TO FARMER

icon

Select Farmers...



MEDICAL EXAMINATION

SERVICE MODE

Complete

Schedule

SERVICE TYPE

Disease Control



MISSION DATE

TIME

(DATE)

(TIME)

PRIORITY PROTOCOL

LOW

MEDIUM

HIGH

FINDINGS / DIAGNOSIS

Describe clinical findings / diagnosis...

TREATMENT ACTION

e.g Wound Cleaning, Injection...

MEDICINE & DOSAGE

← Walk-in Registration

icon

(text)

PERSONAL DETAILS

FIRST NAME*

LAST NAME*

PERSONAL DETAILS

PHONE NUMBER

EMAIL (OPTION)

BARANGAY

MUNICIPALITY

SELECT



OTON

Register Farmer

← Register Livestock

icon

(text)

OWNER / CLIENT SELECTION

icon

Select Client / Owner...



ANIMAL IDENTITY



ADD PHOTO

EAR TAG

TAG

GENERATE TAG

BRAND

OPTIONAL

SPECIES

Beef Cattle ▾

BREED

Select Breed ▾

COLOR

Select Color ▾

DATE OF BIRTH

DATE

Register Livestock

← **Pregnancy Check**

icon

(text)

OWNER / CLIENT

icon

Select Client.

▼

LIVESTOCK ANIMAL

icon

Select Animal...

▼

Record Calf Drop

OWNER / CLIENT

icon

Select Client.



icon

(text)

icon

Greetings

(text)

icon

Search former by name or email...

- All
- Place
- Place
- Place
- Place

(Name)

(Email)

(Address)

✓

>

(Name)

(Email)

(Address)

✓

>

(Name)

(Email)

(Address)

✓

>

(Name)

(Email)

(Address)

✓

>

Quick Actions



icon

Field Calendar

icon

Photo Note

icon

Heatmap

icon

Field Reports

icon

Register Farmer

icon

Ask Moowie

← Field Calendar



(Calendar)

Today's Agenda

0 Tasks



No visits Scheduled

Enjoy your quiet day!



Photo Notes

1 Total Logs



(TEXT)

(ADDRESS) (DATE)

(NOTES)

(NAME)

← Animal Hub

Register

icon

(text)

icon

Search former by name or email...



(Tag)
(Breed)
(Owner)



(Tag)
(Breed)
(Owner)



(Tag)
(Breed)
(Owner)



(Tag)
(Breed)
(Owner)



icon

HOME

icon

USERS



icon

ANIMALS

icon

LEDGER



Service Ledger





(text)



Search by Farmer, Animal Ear Tag, Status...

- All
- A.I
- Pregnancy
- Calving
- Health Log

SHOWING 3 OF 3 RECORDS

EXPORT



(SERVICES)
(FARMER)
(STATUS)

(DATE)
>



(SERVICES)
(FARMER)
(STATUS)

(DATE)
>



(SERVICES)
(FARMER)
(STATUS)

(DATE)
>



Account

icon



(NAME)
(PLACE)

(0)

VISITS

(0)

SUCCESS

(0)

RATING

PERSONAL WORKSPACE

icon

Email & Security



icon

Service Schedule



icon

Notifications



PREFERENCES

icon

App Theme

Light



icon

Privacy Policy



icon

HOME

icon

USERS



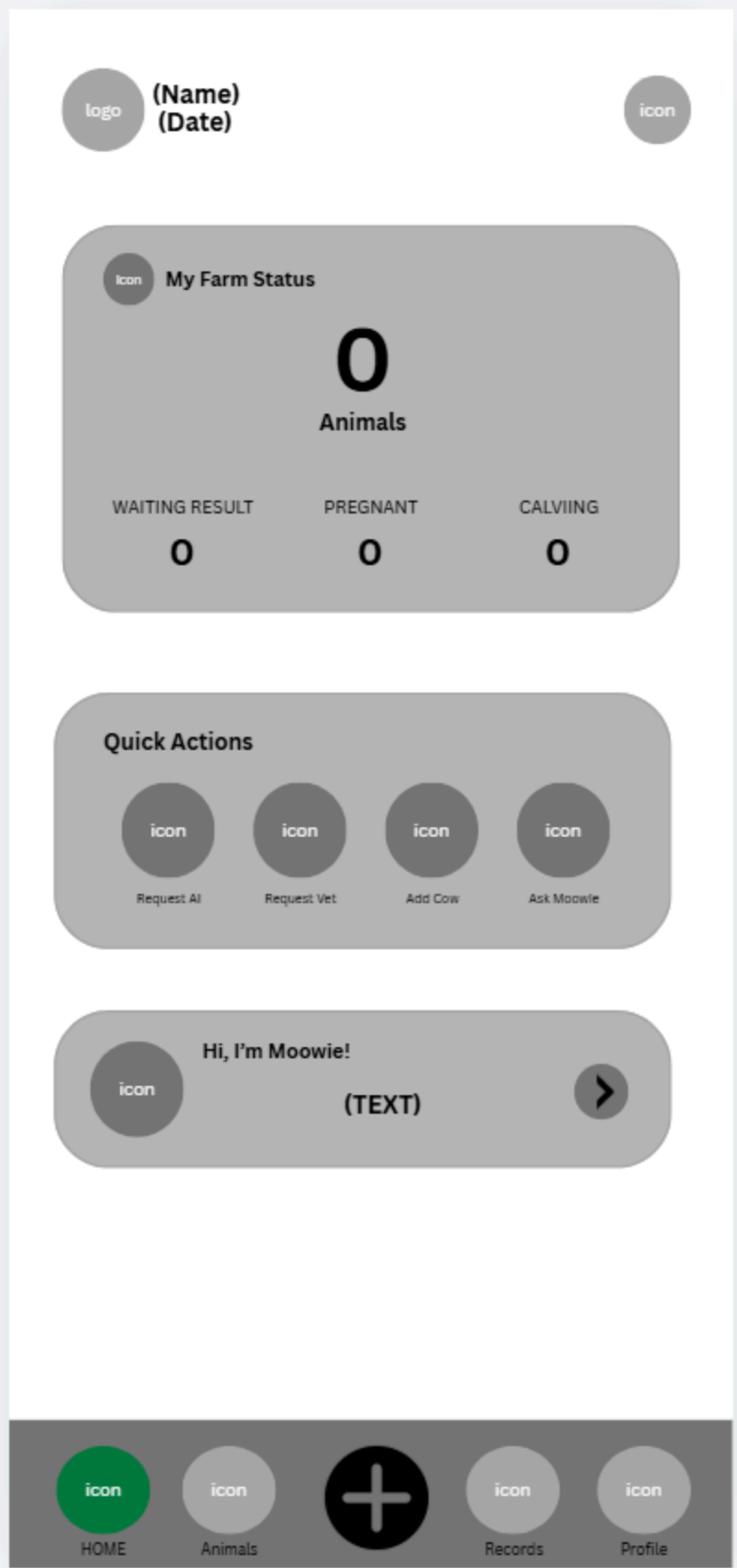
icon

ANIMALS

icon

LEDGER

FARMERS



AI Outcome Check

+more

(TEXT)

(DATE)

Yes, Pregnant!

No, Re-heat

Upcoming Visits

View All

icon

Health Check

(SCHEDULE)

(TECHNICIAN)

IN PROGRESS

CANCEL

Heat & Breeding Alerts

0 - Upcoming Calving

(DATE)

0 - Upcoming Calving

(DATE)

0 - Upcoming Calving

(DATE)



HOME



Animals



Records



Profile

← Request AI Service

YOUR INFORMATION

icon (FULL NAME)

icon (ADDRESS)

SELECT ANIMAL

Tap to choose an animal



PREFERRED VISIT DATE/TIME

DATE

TIME

ATTACH PHOTO (OPTIONAL)



Tap to attach a photo

of the animal in heat

COMMENT / NOTES

NOTES

Submit AI Request

← Report Animal issue



Moowie Support

TEXT

YOUR INFORMATION



(FULL NAME)



(ADDRESS)

AFFECTED ANIMAL

Tap to choose an animal



REQUEST TYPE

Disease / Infection



URGENCY LEVEL

LOW

MEDIUM

URGENT

PREFERRED VISIT DATE/TIME

DATE

TIME

← My Animals



(DATE)

Good evening, Lloyd

 Moowie
(text)

 Search by ID or breed....

 (CODE NAME)
(BREED) >

 (CODE NAME)
(BREED) >

 (CODE NAME)
(BREED) >

 (CODE NAME)
(BREED) >



HOME



Animals



Records



Profile

My Animals

(DATE)

Good evening, Lloyd



Moowie

(text)

EAR TAG

TAG#

GENERATE TAG

SPEICIES

Select

BREED

Select

COLOR

Select

BRAND

OPTIONAL

GENDER

Select

BIRTH DATE

Select

Add to my Farm



HOME



Animals



Records



Profile

Quick Actions



icon

Request AI

icon

Report Issue

icon

My Requests

icon

Add Animal

icon

Disease Map

icon

Ask Moowie



Service Hub

TRACK AND MANAGE YOUR REQUESTS

ALL

PENDING

IN-PROGRESS

REQUESTS

icon

Request AI

Artificial Breeding

icon

Request Vet

Health Checkup

REQUEST ACTIVITY

icon

AI Insemination

(Date)

DONE

icon

(TEXT)
(BREED)

ATTEMPT
#0

ANIMAL DETAILS >

icon

Health Checkup

(Date)

RESOLVED

icon

(TEXT)
(BREED)

ANIMAL DETAILS >

Reports Center



icon

Moowie

(text)

Breeding Cycles

Activity Feed

icon

Upcoming Calving

(TAG)

(DATE)

OVERDUE BY 6 DAYS

>

icon

Heat Watch

(TAG)

(DATE)

1 DAY LEFT

>

icon

Upcoming Calving

(TAG)

(DATE)

10 DAYS LEFT

>

icon

Preg-Check Due

(TAG)

(DATE)

13 DAYS LEFT

>

Reports Center

icon

icon

Moowie

(text)

Breeding Cycles

Activity Feed

0

TOTAL

0

BREEDING

0

HEALTH

0

CALVING

icon

(DATE)

Calving recorded

(Ease)

[View Animal Details >](#)

icon

(DATE)

Calving recorded

(Ease)

[View Animal Details >](#)

icon

HOME

icon

Animals



icon

Records

icon

Profile



(NAME)

REGISTERED FARMER

(0)

TOTAL COWS

(0)

ACTIVE CASES

(0)

PREGNANT

ACCOUNT DETAILS

icon

EMAIL ADDRESS



icon

PHONE NUMBER



icon

FARM ADDRESS



SYSTEM & SUPPORT

icon

Theme Mode

Light >

icon

App Settings



icon

Help Center



icon

Sign Out



icon

HOME

icon

ANIMALS



icon

RECORDS

icon

PROFILE

3.13 Network Plan: Technical Architecture

The BreedSmart system utilizes a Cloud-Synchronized Client-Server Architecture designed to support efficient communication, centralized data management, and real-time information exchange between farmers, livestock technicians, and the Municipal Agriculture Office of Oton, Iloilo. The architecture was developed to address the existing challenges experienced in manual livestock monitoring, such as delayed reporting, inaccurate record-keeping, paperwork duplication, and difficulties in monitoring breeding and health activities across different barangays.

The network plan enables users to access the system through both mobile and web platforms while ensuring that livestock records remain synchronized, organized, and accessible within the municipal agricultural environment. Through cloud-based infrastructure and centralized database management, the system improves operational efficiency, strengthens communication between stakeholders, and supports accurate agricultural reporting.

3.13.1 Network Infrastructure & Topology

The network infrastructure of BreedSmart follows a multi-layered client-server topology to support secure and reliable data communication throughout the municipality. The architecture is divided into three major layers: the Edge Layer, Transport Layer, and Core Layer.

Edge Layer (Field Access Layer)

The Edge Layer serves as the primary point of interaction between users and the system. Farmers and livestock technicians access the platform through smartphones, tablets, laptops, or desktop computers connected through cellular networks such as 4G and 5G or through available wireless internet connections within Oton, Iloilo.

The mobile application allows farmers to submit insemination requests, report livestock diseases, monitor breeding schedules, and review livestock health records in real time. Livestock technicians use the application during field visits to update insemination procedures, vaccination activities, pregnancy diagnoses, and calving information directly from the farm location. This eliminates the dependence on handwritten records and minimizes delays in updating livestock information.

The web-based dashboard is primarily used by administrators and agricultural personnel within the Municipal Agriculture Office. Through this interface, authorized personnel can monitor livestock records, review technician activities, validate reports, and generate official documentation for submission to the Provincial Agriculture Office.

Transport Layer (Communication and Data Transmission Layer)

The Transport Layer is responsible for transmitting information securely between client devices and the cloud server. BreedSmart utilizes RESTful API architecture combined with HTTPS encryption protocols to ensure safe and reliable communication across the network.

Whenever users submit livestock information, requests, or health updates, the data is securely transmitted to the backend server for processing and storage. The use of encrypted communication protects sensitive agricultural information such as livestock ownership records, insemination histories, vaccination schedules, and reproductive monitoring data from unauthorized access or data interception.

This layer also ensures stable synchronization between the mobile application, web dashboard, and centralized database. By using API-based communication, the system allows real-time updating of livestock records, enabling technicians and administrators to immediately access newly submitted information without requiring manual consolidation of reports.

Core Layer (Cloud Server and Database Layer)

The Core Layer functions as the central processing and storage component of the BreedSmart system. The backend infrastructure is developed using Express.js and Node.js, while MongoDB Atlas serves as the cloud-hosted database management system.

The cloud server processes all system operations, including user authentication, livestock management, insemination scheduling, notification handling, pregnancy monitoring, and report generation. It also handles the automated logic used for breed-specific monitoring and date calculations.

MongoDB Atlas stores all livestock-related information, including animal profiles, insemination records, pregnancy data, vaccination histories, health reports, treatment records, calving information, and administrative logs. The use of a cloud database ensures centralized data storage, allowing authorized users to access updated records regardless of their location.

The cloud-based architecture also improves scalability, allowing the system to accommodate increasing numbers of users, livestock records, and agricultural transactions as the municipality continues to expand its digital livestock management operations.

3.13.2 Operational Synchronization Logic

To maintain absolute data integrity and operational consistency across all connected devices and user tiers, BreedSmart implements a sophisticated, multi-stage synchronization workflow. This mechanism is designed not only to automate routine data entry but also to ensure that the "single source of truth" is continuously maintained across the municipal agricultural ecosystem.

The lifecycle begins at the client-side edge, where farmers submit insemination requests, health concerns, or general livestock status updates through the mobile application. The system

enforces strict input validation protocols at this point, capturing essential metadata—such as unique animal identification, breed-specific reproductive data, symptoms, preferred service windows, and multimedia documentation like symptom images. For field-based livestock technicians, the application facilitates rapid data encoding during farm visits, covering critical inputs such as vaccination logs, pregnancy diagnostic outcomes, insemination success metrics, and calving details. By digitizing these inputs at the point of origin, the system significantly reduces the "reporting lag" and transcription errors characteristic of manual recording.

Technicians may also directly encode field data during farm visits, including vaccination records, pregnancy diagnoses, insemination outcomes, and calving details.

Upon transmission, the cloud-based server acts as the central validation engine. Before any record is committed to the database, the server-side logic subjects the incoming data to rigorous consistency checks. The system dynamically identifies the breed—such as Brahman, Holstein, or Bisaya—and applies context-aware business logic to cross-reference the animal's reproductive history. This server-side intelligence automatically computes and schedules critical milestones, including:

Automated calculations are then generated for important livestock management schedules, including:

- Reheating or estrus return schedules
- Expected insemination intervals
- Pregnancy monitoring timelines
- Vaccination follow-up schedules
- Expected calf drop dates

By centralizing these computations, the system removes the variability of manual technician estimation, ensuring that every schedule is mathematically precise and aligned with biological breed constants.

Once validated, updated records are immediately propagated across the BreedSmart ecosystem. The system leverages WebSocket protocols to push real-time notifications to relevant stakeholders, ensuring that data availability is synchronized globally.

- Farmers through the mobile application
- Livestock technicians for scheduling and field operations
- Administrators through the web dashboard

This automated flow creates a seamless information pipeline, significantly reducing administrative overhead, eliminating manual reporting delays, and ensuring that municipal decisions are based on the most current data available.

3.13.3 Reliability and Disaster Recovery

The BreedSmart system is engineered with a multi-layered approach to reliability, explicitly designed to transcend the limitations of traditional, paper-based agricultural record-keeping.

Cloud-Based Reliability and Continuity

Unlike traditional logbooks, which are physical artifacts prone to environmental hazards like moisture, thermal degradation, fire, and biological damage, BreedSmart utilizes the robust infrastructure of MongoDB Atlas. This cloud-native approach ensures that data is not merely stored but preserved within a highly available, globally distributed environment. This design guarantees that authorized agricultural personnel maintain secure, 24/7 access to critical

livestock records, regardless of whether they are working from a municipal office, a remote farm, or a field site, ensuring business continuity despite local facility disruptions.

The cloud environment provides continuous system availability, allowing authorized users to access livestock records anytime and from different locations within the municipality.

Automatic Backup Mechanisms

Data persistence is safeguarded through a regimented automated backup schedule. These systematic backups create point-in-time recovery capabilities, allowing the system to revert to a previous state in the event of accidental deletion, catastrophic hardware failure, or unintended system interruptions. This robust backup strategy acts as a critical fail-safe for preserving years of accumulated agricultural intelligence, including historical breeding performance, long-term health monitoring, and vaccination trend analysis.

Data Redundancy and High Availability

BreedSmart leverages MongoDB Atlas's distributed clustering model to maintain multiple concurrent copies of the database across different physical nodes. This redundancy ensures that the system architecture remains inherently fault-tolerant; if a single server node encounters technical failure or connectivity loss, the infrastructure automatically fails over to a secondary node. This seamless transition minimizes downtime and ensures that service remains operational for the municipal office and field workers at all times.

Security and Access Control

Security is enforced through a strict role-based access control (RBAC) framework. The system implements granular permissions where farmers, livestock technicians, and administrators interact only with data relevant to their specific roles and authorized jurisdictions. All

communication between the mobile client, the web dashboard, and the central database is secured using industry-standard HTTPS protocols, utilizing TLS encryption for data in transit and encryption at rest. This architecture ensures that sensitive agricultural records, user credentials, and municipal statistics are shielded from unauthorized access, cyber threats, and data breaches.

3.13.4 Network Performance and Scalability

The network architecture of BreedSmart is engineered with a modular design philosophy, ensuring it is prepared for both the current operational needs of the Municipality of Oton and the inevitable growth of digital agricultural demands. By utilizing cloud-native infrastructure, the system decouples processing power from local hardware, allowing it to scale dynamically to handle increased user volume, expanding livestock records, and a higher density of agricultural transactions without performance degradation.

Future-Proofing and Technological Integration

- GPS-based livestock monitoring
- IoT-enabled animal health sensors
- Automated SMS notification systems
- GIS-based livestock mapping
- Expanded reporting and analytics modules

By prioritizing a scalable architectural design, BreedSmart ensures that the investment made in today's system provides a resilient and expandable base, capable of evolving alongside the municipality's broader objectives for agricultural modernization and digital transformation.

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APPENDICES

APPENDIX A: Comprehensive Software Architecture and System Dependency Documentation

The systematic configuration, deployment, and long-term maintenance of the **BreedSmart** platform necessitate the rigorous documentation of all software dependencies, runtime environments, and framework iterations to guarantee peak performance within the municipality of Oton, Iloilo. In strict accordance with the developmental research framework, this section delineates the technical system requirements essential for the execution of the web-based administrative portal and mobile client applications.

The Mobile Application Subsystem is engineered upon a cross-platform React Native codebase, bundled and compiled utilizing the Expo application framework Version 50.0.0+. To ensure optimal compatibility with the standard smartphone assets utilized by local livestock raisers in rural sectors, the application identifies Google Android Version 9.0 (API Level 28) as the minimum deployment baseline. For high-fidelity client rendering, the system further supports devices operating on Apple iOS Version 14.0 or above. Localized data operations on the

client-side utilize the native **AsyncStorage** interface synchronized with **TanStack Query** lifecycle states. This configuration efficiently manages transient memory tasks and the offline transaction queue, enabling resilient field operations even during temporary cellular signal volatility in remote agricultural zones.

The Central Administrative Dashboard handles high-density data aggregation and municipal synchronization within a web-based environment, necessitating specific runtime engines. The web interface leverages React.js built upon a Node.js LTS engine Version 20.x, compiled via Vite automation tools to ensure rapid interface state transitions. The styling architecture utilizes the Tailwind CSS compiler Version 3.4 integrated with NativeWind configurations to render views accurately across mobile and desktop displays. To mitigate visualization discrepancies, office terminals must access the platform through modern V8 JavaScript engine clients, such as Google Chrome Enterprise Version 118+, Mozilla Firefox Extended Support Release Version 115+, or Microsoft Edge Chromium Version 119+.

The Server-Side Cloud Infrastructure is hosted on an Ubuntu Server Enterprise Linux LTS Build 22.04 environment, which facilitates main system services and processes application requests. The API routing layer implements the Express.js framework Version 4.19+ built upon the Node.js runtime core. This backend configuration processes validation logic, executes algorithmic breeding cycle computations, and interacts with the NoSQL database layer through Mongoose drivers Version 8.x. The centralized storage environment resides on a MongoDB Atlas Cloud Enterprise instance Server Cluster Tier M10+, ensuring real-time data synchronization and protection for critical agricultural documentation. External security is managed via Clerk Identity Protection services Version 4.x, while the Google Gemini Pro API protocol powers the sophisticated multi-language processing features of the Moowie AI module.

APPENDIX B: Detailed Engineering Specifications for Physical Hardware Components and Devices

To sustain day-to-day operations and prevent database bottlenecks during intensive administrative tasks, the hardware infrastructure must adhere to precise technical specifications. This section describes the minimum and recommended physical equipment requirements necessary for local government workstations and mobile field operations.

The Office Server and Desktop Station is designed to handle heavy administrative workloads, including the compilation of municipal summaries, rendering of complex data tables, and the display of tracking heatmaps for the Oton Agriculture Office. The mandated configuration includes an Intel Core i5 or AMD Ryzen 5 Hexa-Core 64-bit multi-threaded microprocessor with a baseline frequency of 3.20 GHz. The motherboard assembly must accommodate 16 Gigabytes (GB) of DDR4 dual-channel random-access memory (RAM) to support simultaneous server processes. Persistent data storage is facilitated by a 512 Gigabyte (GB) NVMe M.2 Solid-State Disk Drive (SSD) to ensure rapid read-write cycles during large-scale database queries. Visual output necessitates an LED backlit desktop terminal with a minimum screen diameter of 19.5 inches and a native resolution of 1920 by 1080 pixels for optimal dashboard viewing.

The Mobile Field Smart Device represents the primary physical tool utilized by livestock technicians to record insemination data and update health files directly from the field. The smartphone or tablet hardware must utilize an ARM Cortex Octa-Core system-on-chip (SoC) with a minimum processing speed of 2.0 GHz. System logic requires at least 4 Gigabytes (GB) of LPDDR4X system memory (RAM) to maintain background data synchronization. Internal storage necessitates a minimum of 64 Gigabytes (GB) of non-volatile flash memory for application indices, images, and field records. For registration and health reporting, devices

must feature a 13-Megapixel digital camera sensor with autofocus capabilities, enabling technicians to capture high-fidelity verification photos of alphanumeric ear tags and individual livestock assets.

APPENDIX C: Detailed Network Topology, Communication Protocols, and Infrastructure Architecture

As **BreedSmart** is engineered as a distributed client-server application, its operational throughput depends heavily on the network framework connecting user touchpoints with the cloud server. This section describes the multi-layered communication pipeline designed to ensure data transit security and synchronization reliability across all barangays in Oton, Iloilo.

The Field Access Layer, or Edge Layer, manages the initial connection points where stakeholders interact with the platform via cellular data. Technicians and raisers connect using 4G Long-Term Evolution (LTE) or 5G New Radio (NR) networks operated by regional telecommunication providers. Mobile data packets are transmitted through the Transport Layer utilizing secure Hypertext Transfer Protocol Secure (HTTPS) formats. This traffic is protected with Transport Layer Security (TLS) Version 1.3 encryption to safeguard sensitive data from interception during transit. For real-time messaging and dashboard updates, the system utilizes the Socket.io WebSocket protocol to maintain persistent connections for automated push notifications.

The Processing and Storage Core Layer facilitates data delivery to the backend API hosted on the cloud server. Upon reaching the server layer, Node.js and Express.js routing middleware validate JSON inputs before executing matching database functions. The storage architecture utilizes MongoDB Atlas, which duplicates records across multiple cloud data centers to ensure backup access during technical anomalies. To maintain optimal system usability, the network

infrastructure requires a minimum upload and download bandwidth of 2.0 Megabits per second (Mbps). Packet latency should be maintained under 150 milliseconds (ms) to ensure the accuracy of data tables and the proper synchronization of offline records.

APPENDIX D: Formal Research Transmittals and Project Administrative Correspondence Documents

This appendix contains official communications, request forms, and authorization documents filed by the research team to secure access to local records and finalize the development study. These transmittals serve as formal records of coordination between the academic institution and the local government office of Oton, Iloilo.

The primary transmittal document is the official Request for Field Access and Pilot Implementation Support, submitted to the Municipal Agricultural Officer of Oton on March 11, 2026. Signed by the project researchers, this document formally defines the goals of the capstone project. It outlines the operational boundaries of the pilot testing phase and requests authorization to interview field staff, inspect physical paper ledgers, and conduct evaluation sessions with local livestock raisers.

The document further provides the agricultural agency with a written data privacy guarantee. It specifies that all information collected during field testing—including livestock tallies, ownership registries, and health files—will adhere to strict encryption protocols. This documentation certifies that the **BreedSmart** platform functions as a secure tool for community development and public service enhancement while maintaining institutional research ethics.

APPENDIX E: Software Quality Testing Protocols and User Evaluation Instrument Records

This section contains the official software quality testing forms, verification checklist templates, and evaluation tools utilized to analyze application performance. These instruments measure the system against software quality benchmarks during user acceptance testing (UAT) cycles.

The primary form template is the User Acceptance Testing (UAT) Sign-Off and System Verification Log, used to collect feedback from participants following each testing iteration. The form categorizes reviewers into three explicit roles: Office Administrators, Livestock Technicians, and Registered Farmers, ensuring a multi-dimensional evaluation of the user experience. The evaluation table delineates multiple testing scenarios, including user registration, animal identity entry, automated gestation computation, and offline synchronization tests.

Beneath these functional tables, the form includes open-ended sections for user commentary and interface suggestions. This feedback is essential for refining features during iterative development cycles. The form concludes with a signed system confirmation statement, verifying that the application workflows satisfy technical objectives and comply with required ISO 25010 software quality benchmarks prior to final deployment.

APPENDIX F: Professional Curriculum Vitae and Technical Experience Profiles of the Researchers

This section presents the formalized professional profiles of the five Information Technology candidates specializing in the System Development Track who facilitated the design, analytical modeling, programmatic execution, and technical documentation of the **BreedSmart** ecosystem for the Municipality of Oton, Iloilo. Each curriculum vitae index delineates academic trajectories,

specialized technical competencies, and the precise structural responsibilities assigned to each researcher throughout the comprehensive project lifecycle at PHINMA University of Iloilo.

Curriculum Vitae — John Lloyd Cabanig (Project Manager)

- **Academic Background Profile:** Undergraduate Candidate, Bachelor of Science in Information Technology (BSIT), College of Information Technology Education, PHINMA University of Iloilo.
- **Specialization Track Focus:** System Development Track, prioritizing project lifecycle synchronization, systems analysis, Agile development frameworks, and MERN stack architectural coordination.
- **Core Systems Competencies:** Agile Scrum methodology, distributed system architectures, cross-functional project scope enforcement, and Node.js/Express.js backend infrastructure.
- **Project Assignment Role:** Project Manager and Backend Associate; structurally responsible for the strategic maintenance of the project Gantt chart, supervising milestone throughput across the research cluster, and providing technical support in the development of backend services utilizing the MERN stack.

Curriculum Vitae — Nelmar Buenafe (Lead UI/UX Designer)

- **Academic Background Profile:** Undergraduate Candidate, Bachelor of Science in Information Technology (BSIT), College of Information Technology Education, PHINMA University of Iloilo.
- **Specialization Track Focus:** System Development Track, concentrating on user interface architectural design, experience mapping for localized agricultural environments, and responsive style compilation.

- **Core Systems Competencies:** Tailwind CSS interface configuration, NativeWind mobile property styling, hierarchical system layout design, and responsive UI prototyping.
- **Project Assignment Role:** Lead UI/UX Interface Layout Architect; structurally responsible for blueprint modeling, designing high-fidelity wireframes for the administrative web portal and mobile client, mapping navigation logic, and evaluating interface metrics against ISO 25010 usability benchmarks.

Curriculum Vitae — Gian Rovik Some (Lead Documentor I)

- **Academic Background Profile:** Undergraduate Candidate, Bachelor of Science in Information Technology (BSIT), College of Information Technology Education, PHINMA University of Iloilo.
- **Specialization Track Focus:** System Development Track, specializing in formal academic writing, research architectural structuring, systematic literature analysis, and manuscript optimization.
- **Core Systems Competencies:** Technical research standards, academic writing methodologies, systematic review protocols, and empirical observation tracking.
- **Project Assignment Role:** Lead Documentor I; structurally responsible for the primary management and synchronization of the core research manuscript, aggregating critical field research datasets, conducting technical reviews of analytical segments, and supervising the foundational documentation lifecycle.

Curriculum Vitae — Justine Balmores (Lead Documentor II)

- **Academic Background Profile:** Undergraduate Candidate, Bachelor of Science in Information Technology (BSIT), College of Information Technology Education, PHINMA University of Iloilo.

- **Specialization Track Focus:** System Development Track, emphasizing collaborative technical writing, peer review frameworks, administrative documentation management, and data synthesis.
- **Core Systems Competencies:** Collaborative research standards, academic correspondence protocols, technical manuscript synthesis, and longitudinal project tracking.
- **Project Assignment Role:** Lead Documentor II; structurally responsible for the collaborative review and management of the research paper in synchronization with Lead Documentor I, including the maintenance of research archives, executing structural manuscript audits, drafting transmittal templates, and organizing final documentation blocks.

Curriculum Vitae — John Arvy Lopez (Lead Programmer)

- **Academic Background Profile:** Undergraduate Candidate, Bachelor of Science in Information Technology (BSIT), College of Information Technology Education, PHINMA University of Iloilo.
- **Specialization Track Focus:** System Development Track, specialized in full-stack engineering, database driver logic, cloud architecture scaling, and asynchronous state synchronization.
- **Core Systems Competencies:** React.js/Vite component composition, Node.js/Express.js API services, React Native mobile client engineering, and cloud-synchronized MongoDB Atlas NoSQL querying.
- **Project Assignment Role:** Lead Programmer; structurally responsible for the execution of core system logic, developing algorithmic breed-specific calculation matrices,

implementing the offline caching engine, establishing API integration layers, and managing secure database connectivity.